

RESEARCH

Open Access



Digital financial inclusion, environmental sustainability and regional economic growth in China: insights from a panel threshold model

Hamdi Becha^{1,2}, Maha Kalai^{1,2}, Saifeddine Houidi^{1,2} and Kamel Helali^{1,2*} 

*Correspondence:
kamel.helali@fsegs.usf.tn

¹ Faculty of Economics
and Management of Sfax,
University of Sfax, Airport Road
Km 4.5, B.P. 1088-3018, Sfax,
Tunisia

² Research Laboratory
in Competitiveness,
Commercial Decisions
and Internationalisation
(CODECI), Sfax, Tunisia

Abstract

As digital technologies rapidly transform global economies, the digital financial inclusion index (DFII) has emerged as a critical driver of economic growth, particularly in developing regions. In the context of China, where regional economic disparities persist, the expansion of digital financial services offers a promising avenue to boost regional development, promote inclusive growth, and enhance environmental sustainability. This study looks into how digital financial inclusion affects regional economic growth (GRP) in China's provinces. It does this by using a panel threshold autoregressive model (PTAR) and a panel smooth transition autoregressive model (PSTAR) to see if the relationship is not always linear. Drawing on data from 31 Chinese provinces between 2003 and 2022, the analysis uncovers significant threshold effects, revealing distinct phases where digital financial inclusion's influence on economic growth intensifies as financial inclusion increases. Furthermore, digital financial services empower small and medium-sized enterprises (SMEs) and individual entrepreneurs to invest in eco-friendly innovations and low-emission technologies. By lowering barriers to funding, digital financial inclusion encourages the adoption of cleaner production processes and efficient resource management, which can significantly decrease emissions and improve air quality. Additionally, enhanced financial literacy and access to information through digital platforms enable consumers to make environmentally conscious choices, further contributing to reduced pollution levels. These findings provide empirical evidence of the transformative role of digital finance in both regional development and environmental sustainability, underscoring the need for policy interventions that enhance financial inclusion to drive economic growth.

Keywords: Digital financial inclusion, Regional economy growth, Panel threshold models, China

JEL Classification: C33, G10, G20, O11, O53

1 Introduction

Digital financial inclusion is one of the fastest-growing and most widely acknowledged concepts in today's financial landscape and has captured significant attention from both policymakers and businesses due to its potential to revolutionize access to financial services. As the digital economy expands, the importance of promoting financial inclusion

through digital platforms has become a central focus, driven by innovation and fostering greater economic participation across diverse sectors and populations (Gabor and Brooks 2017; Duvendack and Mader 2020; Lai et al. 2020). In addition, the digitization of financial services has the potential to minimize unit transaction costs for customers, expand the scope of transactions that can be conducted, expand the range of transactions that can be carried out, and offer low-cost access to financial services to those who are disadvantaged (Ozili 2018; Li and Feng 2020; Suri et al. 2021).

The initiation of the “Plan for Promoting the Development of Inclusive Finance” in 2015 by the Chinese government’s has resulted in a notable expansion of digital financial inclusion in China (Lai et al. 2020). In accordance with the aforementioned plan, the objective of financial inclusion is to facilitate the provision of financial services by financial institutions to all segments of the population within their jurisdictions that require financial products and services. This is to be achieved on the basis of equal opportunity, commercial viability, and cost-effectiveness. Furthermore, the growth in global demand following the pandemic accelerates the development of digital channels, which is particularly evident in the increased interest from governments, businesses, and the general public in fintech solutions.

The scope of inclusive finance services encompasses disadvantaged groups, including farmers, elderly individuals, and economically disadvantaged populations (Liu et al. 2024). Moreover, this concept was officially introduced during the G20 Summit in Hangzhou in 2016 and encompasses all initiatives aimed at fostering inclusive digital technology integration with the financial sector (Liu et al. 2021b).

Additionally, China has become a leading global player in the field of online alternative finance, with a significant increase in transaction volume (Niranjanamurthy et al. 2019). The country is a leader in the digital economy development, which reached \$5.4 trillion in 2020, positioning it as the second largest digital economy in the world, behind the United States. The rapid expansion of information infrastructure and the digital industry fuels this significant growth.

In recent years, China’s digital economy has undergone a period of rapid development, largely due to the utilization of innovative technologies such as big data and cloud computing (Yin et al. 2019). This has led to the emergence of new digital payment platforms, including Alipay and WeChat Pay. According to the China Internet Network Information Center, in June 2019, 633 million Chinese citizens made online payments through Alipay and WeChat, representing a 32.65 million increase compared to December 2018. Consequently, there has been a notable increase in the use of mobile payments in China, and the use of digital channels proves an effective means of meeting the needs of low-income users, and the integration of digital financial services can prove a profitable venture for service providers.

Since the beginning of the twenty-first century, there has been a notable increase in global carbon emissions (Ma et al. 2016). According to the World Bank, global carbon emissions totaled 34.36 billion tons in 2019, marking a substantial rise of more than 40% compared to the levels recorded at the close of the twentieth century (Xiao et al. 2023). Further, China’s continued pursuit of high-quality development necessitates a considerable input of energy, resulting in a corresponding increase in carbon emissions (Zhu and Lin 2004). In February 2021, the State Council of China highlighted the necessity of fully

implementing the concept of ecological civilization and establishing a robust economic system that supports green, low-carbon, and circular development. This is in order to ensure the realization of the goals set out in the “2030 carbon peak” and “2060 carbon neutrality” initiatives (Chen et al. 2021).

Despite growing recognition of the importance of digital financial inclusion (DFI) in promoting economic growth and addressing environmental challenges, limited research has explored its role in balancing these objectives, particularly at the regional level. In addition, the research on the influence path and mechanism of DFI in reducing carbon emissions has been lacking. Among the current body of studies, a consistent consensus regarding the role of digital financial inclusion in environmental sustainability has not yet been reached. To address this gap, this study incorporates DFI and environmental quality into economic growth model in the case of Chinese provinces, focusing on the period 2003–2022. This research employs key environmental indicators—SO₂ (sulfur dioxide), NO_x (nitrogen oxides), and PM_{2.5} (fine particulate matter)—to assess environmental sustainability.

These variables are chosen for their relevance in measuring air pollution and their direct impact on public health, ecological well-being, and economic growth. The evolution of key air pollutants in China—SO₂, NO_x, and PM_{2.5}—reflects the country’s industrial growth and efforts to combat environmental degradation. SO₂ emissions peaked in the early 2000s but declined significantly after 2006 due to stricter regulations and desulfurization technologies. NO_x emissions rose until 2012, driven by increased vehicle usage and industrial activity, but began to decrease following the 2013 Action Plan for Air Pollution Prevention and Control. PM_{2.5} levels, a major health concern, reached critical levels by the early 2010s, prompting the National Air Pollution Control Plan.

This study aims to examine the non-linear impact of digital financial inclusion on regional economic growth across China’s provinces during the period 2003–2022. To achieve this, the analysis employs the panel threshold autoregressive model (PTAR) proposed by Hansen (1999) and the panel smooth transition autoregressive model (PSTAR) developed by González et al. (2005) and González et al. (2017). These models are used to explore the varying effects of DFI across different thresholds and regimes, shedding light on its role in shaping regional economic dynamics.

1. How do variability in digital financial inclusion impact regional economic growth across China’s provinces from?
2. Is there a favorable influence of using digital financial inclusion on environmental sustainability in China’s provinces?
3. How do regional disparities in digital financial inclusion influence the effectiveness of environmental sustainability efforts and economic growth within China’s provinces?

This study makes several key contributions to the literature on digital financial inclusion, environmental sustainability, and regional economic growth in China. First, it provides a comprehensive analysis of the impact of digital financial inclusion on regional economic growth, leveraging data from China’s provinces over nearly two decades and clarifying the extent to which digital financial services contribute to economic growth across different regions, highlighting the role of digital inclusion in fostering economic

development. Second, it explores the interplay between digital financial inclusion and environmental sustainability, providing new insights into how digital financial services can support or hinder efforts to reduce carbon emissions and assessing whether provinces with higher levels of digital financial inclusion have been more successful in achieving sustainable development goals. This contribution is particularly timely given China's ambitious targets for carbon neutrality by 2060 and the increasing global emphasis on green finance.

This paper organizes the remaining sections as follows: Sect. 2 presents the literature review and the development of the hypotheses. Section 3 describes the data used and the estimation methodology. Section 4 presents the empirical results and statistical interpretations. Section 5 presents a discussion of the results. Section 6 focuses on the conclusions and main recommendations.

2 Literature review

2.1 Digital financial inclusion–economic growth nexus

The advent of the digital age has brought a series of advanced technologies to the financial industry, such as big data, artificial intelligence, the Internet of Things, and 5G, and the new economic business model of digital finance has come into being (Banna and Alam 2021; Beck et al. 2018). Therefore, digital finance refers to the use of digital technology to innovate financial products, such as mobile payment, Internet finance, Internet insurance, P2P, and Internet crowd funding, to reach the vulnerable groups excluded from financial services to enjoy affordable financial services by reducing the information asymmetry between lenders and borrowers and transaction costs (Aziz and Naima 2021; Liu et al. 2021a, b; Ozili 2018).

According to Demirgüç-Kunt (2017), financial inclusion refers to the process that enables the broader population and businesses to access and utilize formal financial services more effectively. Ensuring that individuals and enterprises can access and make use of these services is a core aim of financial inclusion. Furthermore, as Rajan (2009) emphasizes, financial inclusion can offer suitable financial services to vulnerable groups, particularly those with low income. On the other hand, Bruhn and Love (2014) defined financial inclusion “as having an account with a recognized financial institution, which individuals and businesses can use for regular activities such as saving, borrowing, insurance, and payments.” Consequently, the authors argued that financial inclusion should be a central element to any nation's strategy for economic growth, due to the fact that it provides disadvantaged communities with enhanced income and employment prospects.

Digital finance growth has opened up many opportunities to lower financial transaction costs, meet emerging user demand, and expand the accessibility of financial services. The adoption of cutting-edge technologies such as information technology, big data, and cloud computing has driven this progress (Yu et al. 2020).

However, China's financial system remains underdeveloped, hindering its sustainable economic growth (Lardy 2008; Huang and Wang 2011). Banks that rely on credit and collateral-based operations traditionally dominate the system. Moreover, inefficiencies and distortions plague this traditional model. These include limited competition among commercial banks, leading to inflated profits and a lack of sufficient financial services

in rural areas. Additionally, financing constraints, such as high costs and limited access to funding, have hampered China's sustainable economic growth. To address these challenges, digital finance has rapidly emerged as a new financial model in the country (Arjunwadkar 2018).

The relationship between financial inclusion and economic growth is deeply rooted in theoretical frameworks, such as Levine's (2005) Theory of Finance and Growth, the System Theory of Financial Inclusion, and the Diffusion Theory of Innovation. Levine's theory highlights how financial institutions and intermediaries enhance resource allocation by reducing information costs, thereby promoting economic growth. Enhanced financial access enables deposits from newly banked individuals to fund loans for investment, production, and consumption, directly boosting economic output. Ozili (2018) and Ozili et al. (2023) extend this by emphasizing how financial accessibility, including digital financial inclusion, integrates individuals and firms into the formal financial system, driving broader economic growth.

The System Theory of Financial Inclusion suggests that improvements in financial inclusion positively influence various economic sub-systems, including financial, economic, and social structures, ultimately benefiting overall economic performance. Financial inclusion indicators, both traditional and digital, play a critical role in this process, justifying their use in developing a comprehensive financial inclusion index to measure its effect on economic growth. The Diffusion Theory of Innovation underscores the role of technological advancements, particularly in mobile and internet technologies, in enhancing financial inclusion. These innovations reduce the cost of providing tailored financial services, increasing accessibility, and fostering economic activities. In regions with significant ICT adoption, the diffusion of mobile and internet technologies has improved productivity, living standards, and access to essential social services, highlighting their role in financial inclusion and economic growth.

Prior examinations of Chinese provinces have demonstrated that diversified digital financial products can positively impact regional economic growth in remote and underdeveloped western regions (Liu et al. 2021a, b; Ahmad et al. 2021; Sun and Tang 2022). Additionally, these products have been shown to reduce regional economic quality heterogeneities (Lv et al. 2022). In particular, the inclusion of digital financial services can facilitate the application of agricultural techniques by farmers, which in turn benefits the growth of the rural economy (Ma and Li 2021; Zhou et al. 2022; Xu and Wang 2023). Nevertheless, some studies have expressed concern that digital financial inclusion may merely redirect financial activities from the informal sector without actually assisting financially excluded areas or individuals (Yu and Wang 2021; Lai et al. 2022).

Rapih and Wahyono (2023) examined the impact of these two forms of financial inclusion on economic growth across 130 developed and developing countries during 2014 and 2017 by using advanced techniques such as sample splitting and threshold estimation. The authors' results highlighted two key findings. First, traditional financial inclusion demonstrated a more substantial impact on economic growth in countries with lower inclusion levels, emphasizing its role in addressing foundational financial access barriers. Second, digital financial inclusion exhibited a stronger positive effect in countries with higher levels of digital adoption, underscoring its importance in leveraging technology to drive economic development. These findings

provide valuable insights into how financial inclusion policies should be tailored based on a country's level of financial development and digital infrastructure.

Xi and Wang (2023) explore the impact of digital financial inclusion (DFI) on the quality of economic growth in China, using provincial panel data and applying methods such as the panel fixed effects model, mediation effect model, panel threshold model, and spatial Durbin model. Their findings indicate that digital financial inclusion positively influences economic growth quality, particularly in the eastern provinces and regions with higher levels of marketization, and highlight that digital financial inclusion stimulates entrepreneurial activity, leading to an increasing non-linear positive effect on economic growth quality in recent years. Furthermore, digital financial inclusion benefits neighboring regions by promoting inter-regional economic growth quality through spillover effects. The study underscores the importance of expanding the coverage and deepening the usage of DFI to enhance the overall quality of economic growth in China.

Li (2024) investigating the non-spatial and spatial effects of digital financial inclusion on urban and rural economic development for 31 provinces from 2011 to 2019 and found that digital financial inclusion has positively localized effects on rural economic development. Additionally, the spatial spillover effect of digital financial inclusion in rural regions is more profound than that of traditional finance. Furthermore, digital financial inclusion has spatial spillover effects in narrowing urban–rural inequality that traditional finance lacks.

Basnayake et al. (2024) examine the impact of digital financial inclusion on economic growth in 30 Asia-Pacific countries based on panel data for 2014, 2017, and 2021 using fixed-effect and panel threshold estimation techniques to examine the linear and non-linear nexus between DFI and economic growth using country panel data from the years 2014, 2017, and 2021. The results have confirmed that digital financial inclusion significantly promotes economic growth in Asia-Pacific countries and that the development of digital financial inclusion advances economic growth in high-income and upper-middle-income countries. However, though digital financial inclusion is low in low-income countries, the increasing effect of digital financial inclusion has a significant contribution to economic growth compared to high-inclusive countries, and the panel threshold regression model statistically proves that DFI has different roles under low-inclusive and high-inclusive regimes in enhancing economic growth.

In a recent study, Ben Abdallah et al. (2024) showed a non-linear relationship between digital financial inclusion and economic growth, highlighting the presence of threshold levels that influence digital financial inclusion effectiveness in stimulating economic progress. Their findings reveal that while digital financial inclusion initially promotes economic growth significantly, its benefits tend to decrease once specific thresholds are surpassed, emphasizing the importance of balancing the growth of digital financial inclusion with economic development.

H₁: Digital financial inclusion has a non-linear effect on regional economic growth.

The non-linear effect of digital financial inclusion on regional economic growth often involves an initial phase of rapid growth followed by diminishing returns. In

regions with low financial inclusion, digital tools can significantly boost economic activities and entrepreneurship. However, as financial inclusion increases, additional improvements may have less impact.

2.2 Digital financial inclusion–environmental sustainability nexus

The incorporation of digital finance has the potential to offer a range of benefits that could improve environmental outcomes (Chang et al. 2023). Firstly, digital finance can reduce transaction costs associated with traditional banking, making it more accessible and affordable for both individuals and businesses to use financial services and encourage investments in sustainable infrastructure and renewable energy, contributing to lower pollution levels. Secondly, it can support financial flows, ensure spending aligns with environmental sustainability, thereby improving transparency and accountability, and foster international cooperation, enabling countries to work together in tackling global environmental challenges.

Finance was included in environmental-related research, which was also initially developed around the EKC hypothesis (Dinda 2004; Jalil and Feridun 2011). With the deepening research, its content gradually reduces the influence of the EKC hypothesis and pays extra attention to the relationship between finance and the environment. Currently, two main viewpoints exist on the correlation between finance and the environment. The first viewpoint is that financial development contributes to enhancing environmental quality. Indeed, finance can enhance environmental quality by promoting technological progress, adjusting the industry structure, alleviating the shortage of environmental protection funds for enterprises, optimizing the allocation of financial resources, improving the conversion rate of deposits and loans, and promoting green project investment and financing (Hu et al. 2022; Gu et al. 2021; Shahbaz et al. 2013a, b; Tamazian et al. 2009). The second view is that financial development will aggravate environmental pollution. Some scholars believe that financial development increases energy consumption and pollution emissions by easing corporate financing constraints to expand production capacity (Cole and Elliott 2005; Shahbaz et al. 2013a, b) and easing consumer budget constraints to stimulate commodity consumption (Zhang 2011; Sadorsky 2010).

Several studies suggest that digital financial inclusion can indirectly contribute to reducing pollution emissions through pathways such as upgrading the industrial structure, fostering innovation, and encouraging entrepreneurial activities. Specifically, the role of digital financial inclusion in industrial structure upgrading can be understood in three dimensions. First, digital financial inclusion increases the availability and effectiveness of financial services. This helps people who do not have access to them, makes the best use of resources, and directs them to high-productivity areas, which makes it easier to upgrade the industrial structure. Secondly, by driving the growth of the digital economy and supporting the development of logistics, transportation, and other service sectors, digital financial inclusion accelerates industrial transformation through online consumption and structural changes. Thirdly, digital financial inclusion eases budget constraints for low- and middle-income groups, enabling shifts in consumption patterns that further stimulate industrial restructuring. Since service industries typically require less investment and generate lower pollution compared to secondary industries, such transformations help in reducing emissions.

Digital financial inclusion also helps lower emissions through innovation. It does this by making it easier for technology-driven businesses to get outside funding and by fixing problems with misallocating financial resources, which can happen when start-ups do not have enough money or information. This financial inclusion promotes technological advancements within enterprises, thereby reducing pollution emissions (Wen et al. 2020; Xue and Zhang 2022; Li et al. 2022a, b; Lin and Ma 2022, Wang et al. 2023a, b).

An increasing number of empirical studies demonstrated the positive effect of digital financial inclusion on environmental performance. Indeed, an increase in access to capital for environmentally focused businesses and renewable energy projects can contribute to a decrease in pollutant emissions. This is supported by studies conducted by Chaudhry et al. (2022) and Liu et al. (2022). Furthermore, digital finance may enhance company ESG performance by fostering green innovation and external scrutiny (Ren et al. 2023a).

An increasing number of empirical studies demonstrated the digital financial inclusion positive effect on environmental performance. Indeed, an increase in access to capital for environmentally focused businesses and renewable energy projects can contribute to a decrease in pollutant emissions. (Chaudhry et al. 2022; Liu et al. 2022). Furthermore, digital finance may enhance company ESG performance by fostering green innovation and external scrutiny (Ren et al. 2023a).

Ren et al. (2023b) suggest that digital finance can modernize urban industrial infrastructure and reduce pollutant emissions. Moreover, expanding financial inclusion may help alleviate energy poverty, further contributing to lower emissions (Wang et al. 2023a, b). Additionally, stable and supportive policy environments are crucial, as climate policy uncertainty can adversely affect business productivity (Ren et al. 2023a, b).

Finance plays a crucial role in the real economy by delivering essential services and resources that support investment in sustainable projects, encourage the adoption of clean technologies, and promote responsible business practices, which not only enhance financial inclusion, but also accelerate the shift towards a greener economy.

The novel business model, founded upon digital technology, markedly diminishes both the financial outlay and the operational costs associated with financial services, while simultaneously enhancing service efficiency. Conversely, advances in digital finance have addressed the shortcomings of traditional finance, including issues of attribute, territory stage mismatch, facilitate corporate finance, and support technological innovation (Tang et al. 2020).

Additionally, it offers an opportunity to improve the coordination and integration of financial institutions regarding the dispersed resources within society, resulting in more efficient resource allocation and contributing to the industrial structure optimization and advancement (He and Shen 2021; Chang et al. 2023), a promising avenue for attaining the objective of carbon neutrality.

Cao et al. (2021) asserted in their study the dual benefit of digital financing, namely a reduction in energy consumption and an improvement in environmental performance. Indeed, digital finance can facilitate the growth and reduction of carbon emissions within the context of a green economy through the enhancement

and augmentation of industrial structure (Li et al. 2022a, b) and reduce transaction costs, which can lead to reduced energy consumption, encourage innovative practices online, and enhance carbon emissions monitoring (Shao et al. 2022).

Ozturk and Ullah (2022) conducted an extensive analysis of 42 countries participating in the Belt and Road Initiative (OBRI) from 2007 to 2019. The empirical analysis is based on pooled ordinary least squares (OLS), two-stage least squares (2SLS), and generalized method of moments (GMM) approaches. The empirical finding showed that digital financial inclusion increases economic growth but decreases environmental quality through the surge of CO₂ emissions.

Li and Guo (2024) analyze the impact of digital finance on environmental pollution using balanced panel data from 274 prefecture-level cities in China, spanning 2011 to 2018. Their study examines the direct, mediating, nonlinear, and heterogeneous effects of digital finance on pollution. The findings reveal that digital finance effectively reduces environmental pollution, with its impact increasing over time. Among the components of digital finance, the depth of usage has the strongest influence on reducing pollution, while the level of digitalization has the weakest. The study identifies production technology and industrial structure as key mediators in this relationship, while pollution control technology does not play a significant intermediary role. The pollution reduction effect of digital finance is more pronounced in cities with advanced Internet infrastructure and higher levels of human capital. Additionally, the impact is particularly significant in the western regions of China and in medium-to-small cities.

Wang et al. (2024) investigate the role of digital financial inclusion in fostering the coupling coordination between pollution reduction and low-carbon development in China using panel data from Chinese provinces by applying advanced econometric models to analyze the relationship between digital financial inclusion and environmental sustainability. The findings reveal that digital financial inclusion significantly enhances the coupling coordination between pollution reduction and low-carbon development by improving resource allocation, promoting green technological innovation, and facilitating industrial structure upgrades. Additionally, digital financial inclusion mitigates financial exclusion, empowering low-income groups and small enterprises to adopt environmentally sustainable practices, which underscores the dual benefits of DFI in driving economic growth and achieving environmental objectives.

Zheng et al. (2023) explore the intersection between financial inclusion, the COVID-19 pandemic, and CO₂ emissions in highly polluted nations, highlighting potential opportunities for mitigating the negative impacts of these factors on the environment. Their findings expose that financial inclusion reduces CO₂ emissions in highly polluted economies. Moreover, COVID-19 also reported a negative effect on CO₂ emissions. Thus, it is concluded that financial inclusion and COVID-19 both have improved the quality of the environment in highly polluted economies. Panel quantiles regression also reported similar findings. Financial inclusion impact is found to be significantly negative on CO₂ emissions in all quantiles.

Chaudhry et al. (2022) examine the consequences of financial inclusion on pollution emissions in the OECD economies and report that higher financial inclusion tends to reduce pollution emissions. Researchers have given tremendous attention to the impact of financial inclusion on environmental performance, as financial inclusion is crucial for

financing clean energy ventures. Indeed, financing clean energy projects can improve environmental performance, but ignoring the financial investment effect on the environment can trigger environmental issues (Dou and Li 2022).

Feng et al. (2022) investigates the role of digital financial development in promoting sustainable development in China by reducing the ecological footprint across provinces from 2011 to 2019. The findings reveal that although the provincial ecological footprint per capita increased by an average of 29.15% during the period, digital financial development effectively suppressed it, fostering a balance between economic growth, ecological conservation, and environmental equity. Moreover, bridging the digital divide, strengthening environmental constraints, and enhancing government digital governance were found to amplify the positive effects of digital finance on ecological footprint reduction. Additionally, digital financial development facilitates green technological innovation and enhances the scale and effectiveness of green credit, both of which are critical pathways for sustainable development.

Shabir (2024) explores the effect of financial inclusion on CO₂ emissions in Asia–Pacific Economic Cooperation (APEC) countries from 2004 to 2018, focusing on the moderating roles of technological innovation and economic globalization. Using advanced econometric methods robust to cross-sectional dependence and slope heterogeneity, the augmented mean group (AMG) results show that financial inclusion and renewable energy consumption enhance environmental sustainability by reducing CO₂ emissions. Conversely, economic growth and economic globalization negatively affect environmental quality by increasing CO₂ emissions. Furthermore, the study reveals that financial inclusion supports environmental sustainability through technological innovation, while its interaction with economic globalization negatively impacts environmental quality. The Dumitrescu and Hurlin (D-H) non-causality test identifies unidirectional causality between financial inclusion, renewable energy, and CO₂ emissions, and a feedback relationship among economic globalization, technological innovation, economic growth, and CO₂ emissions.

Bakhsh et al. (2024) examines the potential for promoting environmental sustainability in China by exploring the interplay between financial inclusion, digitalization, and environmental sustainability. Employing a comprehensive methodology that includes quantile-on-quantile regression, Granger causality in quantiles, wavelet analysis, and robustness tests, the research identifies a strong positive relationship among these factors. Their findings that fostering financial inclusion and leveraging advanced technologies are essential for enhancing the environmental sustainability of China's economic growth, provide valuable insights for emerging economies seeking to align economic growth with sustainable development goals through technological innovation and improved financial accessibility.

Jiang et al. (2024) delve into the intricate dynamics between carbon emissions and the digital economy, focusing on 30 Chinese regions over the period 2011 to 2021. Through the application of spatial econometric techniques, the study uncovers that for every 1% increase in the digital economy, carbon emissions decrease by 0.082–0.092%, highlighting the role of digitalization in promoting environmental sustainability. Furthermore, the findings reveal that the growth of the digital economy not only mitigates carbon emissions within a region, but also exerts a positive spillover effect, reducing emissions

in neighboring provinces. This underscores the interconnected nature of regional economies and the potential for digital advancements to generate widespread environmental benefits.

H₂: Digital financial inclusion has a positive effect on environmental sustainability by reducing air pollution.

The integration of digital financial services has the potential to facilitate investments in environmentally sustainable technologies and practices. To illustrate, the expansion of digital financial services facilitates investment in renewable energy solutions or the adoption of green practices by businesses and individuals alike. This transition can diminish reliance on environmentally detrimental energy sources and advance sustainable development, which in turn will contribute to a reduction in air pollution.

3 Methodology

3.1 Theoretical framework

To formalize the relationship between these variables and economic output, the study employs a Cobb and Douglas (1928) production function framework, a well-established model in economic analysis. This framework effectively captures the interplay between traditional inputs—capital and labor—and emerging factors like digital financial inclusion (DFI) and environmental sustainability (ES) as distinct and measurable drivers of growth. The function is designed to quantify how these variables contribute to economic output, with DFI and ES integrated as additional components reflecting their growing relevance in modern economies.

In traditional economic growth models, capital and labor remain the cornerstones of economic output. Capital investments provide the necessary infrastructure to support production activities, while labor represents the human effort driving economic performance. Even with the increasing digitalization of economies, research consistently underscores the continued relevance of these foundational inputs in sustaining economic growth.

The digital financial inclusion index (DFII) plays an increasingly significant role in enhancing economic output, as reflected by its elasticity γ . By utilizing digital platforms, DFII expands access to financial services, enabling entrepreneurship, boosting investments, and improving resource allocation, particularly in underserved areas. It has proven particularly impactful in reducing poverty and supporting small business growth, especially in rural regions where traditional banking services are often unavailable. Higher γ values indicate a stronger influence of DFII on economic growth, suggesting that provinces with more robust digital finance infrastructure experience enhanced economic performance through improved financial accessibility.

Environmental sustainability (ES) also contributes to economic output, represented by elasticity δ . Investments in sustainability, such as renewable energy initiatives and pollution control measures, boost productivity by fostering healthier environments and reducing the adverse impacts of environmental degradation. A cleaner environment improves public health, reduces absenteeism, and lowers healthcare costs, all of which enhance labor productivity. A positive γ indicates that regions embracing sustainable

practices not only achieve environmental benefits, but also experience long-term economic stability and resilience (Eq. 1):

$$GRP_t = AK_t^\alpha L_t^\beta DFII_t^\gamma ES_t^\delta e^{u_t}, \quad (1)$$

where GRP_t is the gross regional product, representing total output in period t ; the constant A is the total factor productivity (TFP), reflecting technological advancements and efficiency; K_t is the capital stock, such as infrastructure and machinery investments; L_t is the labor force, representing human capital; $DFII_t$ is the digital financial inclusion index at time t , representing the availability and use of digital financial services; ES_t is the environmental sustainability, represented by green practices like renewable energy adoption; α , β , γ and δ are the elasticities of output with respect to capital, labor, DFI, and environmental sustainability, respectively.

To simplify the estimation of parameters, we can log-transform the production function, converting it into a log-linear form (Eq. 2):

$$\ln GRP_t = \ln A + \alpha \ln K_t + \beta \ln L_t + \gamma \ln DFII_t + \delta \ln ES_t + u_t \quad (2)$$

This form allows us to empirically estimate each input's contribution to GRP using panel data regression across multiple provinces and time periods. Each coefficient represents the elasticity of GRP with respect to its associated input, making it possible to measure the distinct impacts of DFI and environmental sustainability.

The growth rate of GRP can be decomposed as follows (Eq. 3):

$$g_{GRP} = \alpha g_K + \beta g_L + \gamma g_{DFII} + \delta g_{ES}, \quad (3)$$

where g_{GRP} is the growth rate of GRP, and g_K , g_L , g_{DFII} , g_{ES} are growth rates of capital, labor, DFI, and environmental sustainability, respectively. This formulation provides a clear view of how changes in DFI and environmental sustainability independently drive economic growth. An increase in the growth rates of DFI, or environmental sustainability, assuming positive elasticities γ and δ , will boost the overall growth rate of GRP.

3.2 Data and variables

Using a panel data model, this study aims to examine the impact of digital financial inclusion and air pollution on economic growth in China's provinces. In light of the specifications provided by Ding et al. (2022) and Ben Abdallah et al. (2024), the model can be written as follows:

$$\begin{aligned} \ln GRP_{it} = & \beta_0 + \beta_1 \ln POP_{it} + \beta_2 \ln GFCF_{it} + \beta_3 \ln EII_{it} \\ & + \beta_4 \ln DFII_{it} + \beta_5 \ln SO_{2it} + \beta_6 \ln NOX_{it} \\ & + \beta_7 \ln PM_{2.5it} + \lambda_i + \mu_t + \varepsilon_{it}, \end{aligned} \quad (4)$$

where GRP_{it} is the gross regional product variable for province " i " in period " t ", POP_{it} , $GFCF_{it}$, $DFII_{it}$, SO_{2it} , NOX_{it} , and $PM_{2.5it}$ represent all the independent variables, λ_i captures the province-specific effect, μ_t takes into account the relevant time effect, and ε_{it} is a random error term. $\ln$ represents the natural logarithm. All the variables are summarized and defined in Table 1 with their references.

Table 1 Definition of variables

Variables	Definition	References	Sources
LnGRP	Gross regional product (100 million Yuan)	Ben Abdallah et al. (2024)	NBS
LnPOP	Resident population (year-end) (10,000 persons)	Kalai et al. (2024a, b)	NBS
LnGFCF	Gross fixed capital formation (% of GDP)	Kalai et al. (2024a, b)	NBS
LnEII	Energy industrial investment (100 million Yuan)	Ben Abdallah et al. (2024)	NBS
LnDFII	Digital financial inclusion index	Lui et al. (2024); Ding et al. (2022)	PKU-DFII
LnSO ₂	Sulfur dioxide emission in waste gas (10,000 tons)	Bakhsh et al. (2022)	NBS
LnNOX	Nitrogen oxides emission in waste gas (10,000 tons)	Han et al. (2021)	NBS
LnPM _{2.5}	Average PM _{2.5} per 10 km ²	Hao and Liu (2016)	TAP Data

NBS refers to “National Bank of Statistics”, PKU-DFII refers to “Peking digital financial inclusion index”, TAP data refer to “tracking air pollution”

The variables used in this study are integral to understanding the relationship between economic growth, environmental pollution, and digital financial inclusion in Chinese provinces. The GRP variable represents the gross regional product, measured in 100 million Yuan, and is a key indicator of economic activity at the regional level. It is widely used to assess the economic output of provinces and serves as a proxy for regional economic growth (Ben Abdallah et al. 2024).

The POP variable refers to the resident population, measured at year-end in 10,000 persons, and captures demographic trends within regions. Population size influences economic dynamics by affecting labor supply, demand, and regional market size, thus directly impacting economic growth (Kalai et al. 2024a, b). The GFCF variable represents gross fixed capital formation as a percentage of GDP and is an important measure of investment in physical assets such as infrastructure, machinery, and buildings. High GFCF indicates significant investment in the region’s productive capacity, which can foster economic growth (Kalai et al. 2024a, b).

The EII variable denotes energy industrial investment, measured in 100 million yuan, reflecting the level of investment directed towards energy-related industries. This variable is vital for understanding how investments in energy infrastructure and technology contribute to regional economic performance (Ben Abdallah et al. 2024). The DFII variable is the digital financial inclusion index, which measures the extent to which digital financial services are accessible across different regions. It is an essential variable in this study as it reflects the role of digital financial technologies in facilitating economic growth, especially by providing easier access to credit, investment opportunities, and other financial services that can support green technologies and sustainable practices (Lui et al. 2021a; Ding et al. 2022; Ben Abdallah et al. 2024).

In terms of environmental indicators, the LnSO₂ variable represents the emission of sulfur dioxide in waste gases, measured in 10,000 tons. SO₂ emissions are a significant environmental pollutant, contributing to air pollution and having adverse effects on public health and the economy (Bakhsh et al. 2022). NOX refers to nitrogen oxide emissions in waste gases, also measured in 10,000 tons, and similarly plays a crucial role in assessing air quality and environmental impact, with high levels of NOX contributing to acid rain and respiratory issues (Han et al. 2021).

Finally, PM_{2.5} is the average concentration of fine particulate matter (PM_{2.5}) per 10 km², which is a critical air quality indicator. High concentrations of PM_{2.5} are linked

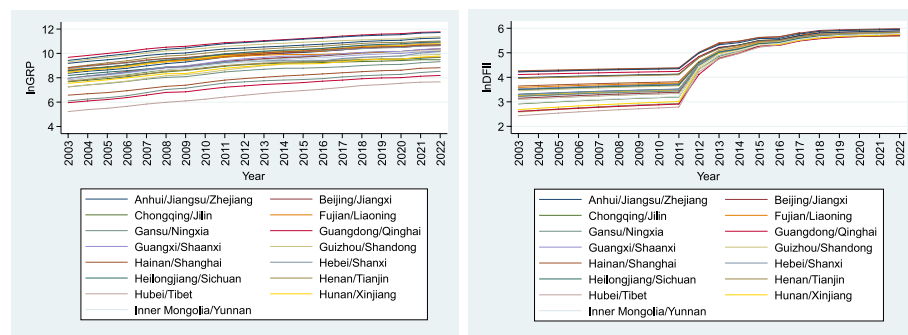


Fig. 1 Trend in GRP and DFIL in Chinese provinces in logarithms

to a variety of health problems and can significantly hinder regional productivity and economic growth (Hao and Liu 2016). These variables together help to form a comprehensive view of how economic activities, financial inclusion, and environmental factors interact in the context of regional development in China.

According to Fig. 1, the economic evolution of gross regional product (GRP) in Chinese provinces between 2003 and 2022 reflects significant regional disparities and growth dynamics influenced by various factors. These disparities arise from several interrelated factors. Coastal provinces like Guangdong and Jiangsu benefit from a diversified, industrialized economy and significant foreign direct investment (FDI) due to better infrastructure and favorable business environments. In contrast, many inland provinces rely heavily on agriculture and natural resources, resulting in lower GRP levels. In addition, rapid urbanization in major cities fosters economic activity, while rural areas face infrastructure deficits that hinder growth.

For this, government policies, such as the “Go West” strategy, have been implemented to boost less developed regions, but their effectiveness varies based on local governance and resource allocation. Additionally, regions with higher educational attainment and skilled labor tend to innovate and grow more rapidly, while geographic location significantly influences economic opportunities.

The emergence of digital financial inclusion in Chinese provinces between 2003 and 2022 has been a transformative economic phenomenon, driven by rapid technological advancements, changing consumer behavior, and supportive government policies. The widespread adoption of smartphones and the internet facilitated the digital financial services rise, allowing individuals and small businesses, especially in rural and underserved areas, to access banking, credit, and investment opportunities that were previously out of reach. In addition, the proliferation of mobile payment platforms like Alipay and WeChat Pay has significantly reduced transaction costs and increased the convenience of financial services, promoting higher levels of financial participation.

Government initiatives, such as the “Internet+” strategy and financial reforms, further supported this trend by encouraging fintech innovations and fostering a regulatory environment conducive to the growth of digital finance. Additionally, the COVID-19 pandemic accelerated the transition to digital financial services, with an increasing number of individuals seeking contactless payment methods and online

banking solutions. However, disparities in digital financial inclusion emerged across provinces, with coastal regions benefiting more from advanced infrastructure and higher digital literacy levels than inland areas, which often face challenges related to internet access and technological adoption.

3.3 Model's specifications

This study, aligned with the research of Bouattour et al. (2023), Bouattour et al. (2024a), and Kalai et al. (2024a, b), expands upon Hansen's (1999) Panel Threshold Autoregressive (PTAR) model, which facilitates the analysis of whether an equation can be divided into multiple regimes based on a threshold variable value, particularly advantageous when (T) is limited and (N) is extensive. In this model, the variable y_{it} is influenced by several non-dynamic interactions. Consequently, Eq. 5 adheres to a two-regime PTAR model, contingent upon the fulfillment of certain criteria:

$$y_{it} = \mu_i + \sum_{j=1}^p \rho_j y_{it-j} + \beta'_1 X_{it} + \beta'_2 X_{it} I(q_{it} > c) + \varepsilon_{it}, \quad (5)$$

where μ_i represents the individual fixed coefficients, ρ_j represents the autoregressive coefficients of the process y_{it} , $I(q_{it} > c)$ represents the indicator function for the transition variable q_{it} , $X_{it} = (X_{it}^1, \dots, X_{it}^k)$ represents k exogenous variables that do not include lagged explanatory variables. The variables in the equation are represented by the vector of coefficients $\beta = (\beta_1, \dots, \beta_k)$ and we suppose that the error term ε_{it} follows a normal distribution with mean 0 and variance σ^2 . The subscript $i = 1, \dots, N$ denotes the specific dimension, whereas the subscript $t = 1, \dots, T$ represents the temporal dimension.

The distinctiveness of this modeling technique resides in the occurrence of a sudden change as a unit might switch from one regime to another at a certain moment. When the change indicator exceeds the threshold variable $q_{it} \leq c$, even minimally, the procedure follows the first regime, using the slope coefficients in a sequential manner. Conversely, if $q_{it} > c$, is greater than c , the process follows the second regime, characterized by the coefficients $(\beta'_1 + \beta'_2)$. Consequently, Eq. 5 can be written as follows in Eq. 6:

$$y_{it} = \begin{cases} \mu_i + \sum_{j=1}^p \rho_j y_{it-j} + \beta'_1 X_{it} + u_{it} & \text{if } q_{it} \leq c \\ \mu_i + \sum_{j=1}^p \rho_j y_{it-j} + (\beta'_1 + \beta'_2) X_{it} + u_{it} & \text{if } q_{it} > c \end{cases} \quad (6)$$

In order to present a procedure for estimating slope and threshold parameters, it is first necessary to develop an r -regime PTAR model.

$$y_{it} = \mu_i + \sum_{j=1}^p \rho_j y_{it-j} + \beta' X_{it}(c) + u_{it}. \quad (7)$$

Equation 6 presents two primary challenges: ensuring consistent estimation of individual effects across different regimes and accurately estimating slope and threshold parameters. Unlike standard time series models, ordinary least squares (OLS) estimation is complicated by the direct linkage between exogenous variables and thresholds.

Therefore, Hansen (1999) suggests an optimal threshold estimator $\hat{c}$ in Eq. 8, aimed at minimizing the residual sum of squares (RSS) as follows:

$$\hat{c} = \arg \min [RSS(c)], \quad (8)$$

where $RSS = \sum_{i=1}^N \sum_{t=1}^T u_{it}^2$.

Furthermore, the residual empirical variance may be obtained using the following equation (Eq. 9):

$$\sigma^2 = \sum_{i=1}^N \sum_{t=1}^T \hat{u}_{it} \hat{u}_{it}^* = \frac{1}{N(T-1)} S(\hat{c}). \quad (9)$$

The initial phase involves conducting a linearity test, which determines whether the null hypothesis is accepted or rejected. Linearity hypothesis rejection triggers two subsequent tests: identifying regime numbers and assessing the slope parameter's significance. The linearity test checks for consistent parameters across regimes, while Eq. 10 tests the hypothesis of no threshold effect relative to Eq. 5:

$$\begin{cases} H_0 : \beta_2' = 0 \\ H_1 : \beta_2' \neq 0 \end{cases}. \quad (10)$$

A preliminary linearity test should be performed in order to assess the coefficient, which should then be treated as being equivalent to its predicted value (Eq. 11):

$$F_1 = \frac{S_0 - S_1(\hat{c})}{\hat{\sigma}^2} \text{ where } \sigma^2 = \frac{1}{N(T-1)} S_1(\hat{c}_1). \quad (11)$$

The residual sums of squares for the linear and non-linear models are represented by S_0 and $S_1(\hat{c}_1)$, respectively.

Moreover, if we assume that we have two regimes ($h = 1$ and $h = 2$), then under these conditions, our basic model (Eq. 4) is written in the PTAR specification as follows (Eq. 12):

$$\begin{aligned} LnGRP_{it} = & \left(\sum_{j=1}^p \rho_{1j}^1 LnGRP_{it-j} + \beta_0^1 + \beta_1^1 LnPOP_{it} + \beta_2^1 LnGFCF_{it} + \beta_3^1 LnEII_{it} \right. \\ & \left. + \beta_4^1 LnDFII_{it} + \beta_5^1 LnSO_{2it} + \beta_6^1 LnNOX_{it} + \beta_7^1 LnPM_{2.5it} + \lambda_{1i} + \mu_{1t} + \varepsilon_{1it} \right) \\ & + \left(\sum_{j=1}^p \rho_{1j}^2 LnGRP_{it-j} + \beta_0^2 + \beta_1^2 LnPOP_{it} + \beta_2^2 LnGFCF_{it} + \beta_3^2 LnEII_{it} \right. \\ & \left. + \beta_4^2 LnDFII_{it} + \beta_5^2 LnSO_{2it} + \beta_6^2 LnNOX_{it} + \beta_7^2 LnPM_{2.5it} + \lambda_{2i} + \mu_{2t} + \varepsilon_{2it} \right) \\ & \times I(LnDFII_{it-1}; c). \end{aligned} \quad (12)$$

The coefficients of the lagged gross regional product ($LnGRP_{it-j}$) at each regime, $h = 1$ or $h = 2$, are ρ_{1j}^1 and ρ_{1j}^2 .

Additionally, this study builds on the work of Gonzalez et al. (2005 and 2017), who developed regime change models that Helali and Kalai (2021), Bouattour et al. (2024b), and Kalai et al. (2024b) improved using panel-smoothed transition autoregressive modeling (PSTAR). This model's contribution is comparable to the results of converting time series from quick transition approaches to soft transition methods.

González et al. (2005) and González et al. (2017) advocate for enhancing the PTAR model with the Panel Smooth Transition Autoregressive (PSTAR) model, incorporating a continuous transition function. In addition, its application is advantageous for multi-section panel data studies, effectively capturing cross-section variability and enabling smooth transition analysis. Therefore, a two-regime PSTAR model (Eq. 13) is written as follows:

$$y_{it} = \mu_i + \sum_{j=1}^p \rho_j y_{it-j} + \beta_1' X_{it} + \beta_2' X_{it} G(q_{it}; \gamma; c) + \varepsilon_{it}. \quad (13)$$

The matrix of k exogenous variables without lagged explanatory variables is presented by $X_{it} = (X_{it}^1, \dots, X_{it}^k)$ and $\beta = (\beta_1, \dots, \beta_k)$ is the vector of their coefficient. $G(q_{it}; \gamma; c)$ shows the transition function associated with the transition variable q_{it} , the threshold parameter c , and the smoothing coefficient γ . μ_i represents the fixed coefficient vector, while, ρ_j represents the process autoregressive coefficients y_{it} , assumedly $\varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$. Each dimension is represented by $i = 1, \dots, N$, whereas temporal dimension is represented by $t = 1, \dots, T$.

In PSTAR modeling, the indicator function is modified by a continuous transition function $G(q_{it}; \gamma; c)$, across the interval (0,1). González et al. (2005) advise using a logistic transition function (Eq. 14) with an order indicated as m :

$$G(q_{it}; \gamma; c) = \left[1 + \exp \left(-\gamma \prod_{j=1}^m (q_{it} - c_j) \right) \right]^{-1}, \gamma > 0 \text{ and } c_1 \leq c_2 \leq \dots \leq c_m, \quad (14)$$

where c is an m -dimensional vector of location parameters denoted by $c = (c_1, \dots, c_m)'$. The smoothness of the transitions is determined by the slope parameter γ . For identification purposes, the limitations $\gamma > 0$ and $c_1 \leq c_2 \leq \dots \leq c_m$ are enforced. In practical application, it is typically adequate to take into account $m=1$ or $m=2$, as these values accommodate frequently observed variations in the parameters.

González et al. (2017) suggested replacing $G(q_{it}; \gamma; c)$ with a first-order Taylor expansion. After re-parameterization, this technique produces the following supplementary regression (Eq. 15):

$$y_{it} = \mu_i + \sum_{j=1}^p \rho_j y_{it-j} + \beta_1'^* X_{it} + \beta_2'^* X_{it} G(q_{it}; \gamma; c) + \varepsilon_{it}^*. \quad (15)$$

The parameters, $\beta_1'^*$ are directly proportional to the slope parameter γ and are calculated by the equation $\varepsilon_{it}^* = \varepsilon_{it} + R_1 \beta_1'^* X_{it}$, where R_1 represents the Taylor expansion remainder.

Prior to the PSTAR model estimation, it is crucial to test linearity, which determines the statistical significance of the regime change impact. Testing $H_0 : \gamma = 0$ is similar to testing the null hypothesis $H_0^* : \beta_2'^* = 0$. The Wald test (Eq. 16), Fisher test (Eq. 17), and likelihood ratio test (Eq. 18) can be used for this purpose. The appropriate statistics for these tests are specified as follows:

$$LM = TN(RSS_0 - RSS_1)/RSS_0, \quad (16)$$

$$LR = -2[\log(RSS_0) - \log(RSS_1)], \quad (17)$$

$$LMF = [TN(RSS_0 - RSS_1)/K]/[RSS_0/(TN - N - K)]. \quad (18)$$

In linear panels with individual effects, RSS_0 represents the residual sum of squares. In non-linear panels with two regimes, RSS_1 refers to the residual sum of squares with are K degrees of freedom, where T represents the number of observations per country and N denotes countries number.

Moreover, if we assume that we have two regimes ($h = 1$ and $h = 2$), then under these conditions, our basic model (Eq. 4) is written in the PSTAR specification as follows (Eq. 19):

$$\begin{aligned} LnGRP_{it} = & \left(\sum_{j=1}^p \rho_{1j}^1 LnGRP_{it-j} + \beta_0^1 + \beta_1^1 LnPOP_{it} + \beta_2^1 LnGFCF_{it} + \beta_3^1 LnEI_{it} \right. \\ & \left. + \beta_4^1 LnDFII_{it} + \beta_5^1 LnSO_{2it} + \beta_6^1 LnNOX_{it} + \beta_7^1 LnPM_{2.5it} + \lambda_{1i} + \mu_{1t} + \varepsilon_{1it} \right) \\ & + \left(\sum_{j=1}^p \rho_{1j}^2 LnGRP_{it-j} + \beta_0^2 + \beta_1^2 LnPOP_{it} + \beta_2^2 LnGFCF_{it} + \beta_3^2 LnEI_{it} \right. \\ & \left. + \beta_4^2 LnDFII_{it} + \beta_5^2 LnSO_{2it} + \beta_6^2 LnNOX_{it} + \beta_7^2 LnPM_{2.5it} + \lambda_{2i} + \mu_{2t} + \varepsilon_{2it} \right) \\ & \times G(LnDFII_{it-1}; \gamma; c), \end{aligned} \quad (19)$$

where ρ_{1j}^1 and ρ_{1j}^2 are the coefficients of the lagged gross regional product ($LnGRP_{it-j}$) at each regime, $h = 1$ or $h = 2$.

4 Empirical results

4.1 Variables description

Before conducting preliminary tests such as homogeneity tests, cross-dependence tests, unit root tests, and cointegration tests, we present the descriptive statistics for the log-transformed variables. The analysis of Table 2 reveals that most of the variables exhibit left skewness (skewness < 0) and are leptokurtic (kurtosis > 0). Additionally, the Jarque

Table 2 Descriptive analysis

Description	LnGRP	LnPOP	LnGFCF	LnEI	LnDFII	LnSO ₂	LnNOX	LnPM2.5
Mean	9.292	8.102	8.886	6.232	4.568	3.278	3.817	3.632
Median	9.429	8.248	8.987	6.384	4.849	3.708	3.919	3.612
Standard deviation	1.192	0.853	1.236	0.980	1.068	1.540	0.895	0.465
Minimum	5.225	5.605	4.897	2.302	2.431	-2.316	1.226	2.140
Maximum	11.768	9.448	11.057	8.227	5.984	5.299	5.212	4.722
Skewness	-0.623	-0.975	-0.532	-0.809	-0.275	-1.420	-0.830	0.092
Kurtosis	3.334	3.527	2.920	4.005	1.542	4.955	3.430	2.540
Jarque-Bera (JB) test	43.06	105.6	29.48	93.91	62.74	307.3	76.09	6.342
JB probability	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.042
Born-Breitung (BB) test	983.86	18.32	327.97	141.98	305.73	105.34	108.57	106.64
BB probability	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

JB refers to Jarque and Bera (1987) test; BB refers to Born and Breitung (2016) test

Table 3 Cross-dependence tests

Variables	Value	Probability	Decision
LnGRP	95.97	0.000	Dependence
LnPOP	36.02	0.000	Dependence
LnGFCF	90.79	0.000	Dependence
LnEI	77.03	0.000	Dependence
LnDFII	96.31	0.000	Dependence
LnSO ₂	85.59	0.000	Dependence
LnNOX	84.78	0.000	Dependence
LnPM2.5	81.96	0.000	Dependence

and Bera (1987) test rejects the null hypothesis of normality for the variables, and the Born and Breitung (2016) examination rejects the null hypothesis of no autocorrelation, with all probabilities being below 1%.

The variable “LnGRP” has an overall mean of 9.292, a median of 9.429, and a standard deviation of 1.192. The skewness value of -0.623 indicates a left-skewed distribution, while a kurtosis value of 3.334 indicates leptokurtic characteristics. The data range from 5.225 to 11.768, and the distribution rejects the null hypotheses of normality and absence of serial autocorrelation at the 1% significance level.

For the variable “LnDFII”, the mean is 4.568, with a standard deviation of 1.068, indicating considerable variability. The 620 observations range from 2.431 to 5.984, with a notable concentration around 4.849. This distribution is asymmetrical, showing left skewness (− 0.275) and leptokurtic characteristics (kurtosis=1.542). Similar to LnGRP, this distribution also rejects the null hypotheses of normality and no serial autocorrelation at the 1% significance level.

Table 4 Slope homogeneity test

Statistics	Value	Probability
Delta	11.372	0.000
Delta adjusted	15.334	0.000

Table 5 Second-generation unit root test

Models	LnGRP	LnPOP	LnGFCF	LnEI	LnDFII	LnSO ₂	LnNOX	LnPM2.5
Panel A: in level								
Constant	− 2.025	− 2.044	− 2.838***	− 2.355***	− 4.357***	− 2.407***	− 1.528	− 2.906***
Constant and Trend	− 2.168	− 0.990	− 2.428	− 2.986***	− 4.397***	− 3.232***	− 2.253	− 3.287***
Decision	NS	NS	NS	S	S	S	S	S
Panel B: in first difference								
Constant	− 3.297***	− 2.654***	− 3.364***	− 4.085***	− 5.199***	− 4.359***	− 2.856***	− 4.559***
Constant and Trend	− 3.486***	− 3.242***	− 3.526***	− 4.115***	− 5.286***	− 4.885***	− 3.489***	− 4.497***
Decision	S	S	S	S	S	S	S	S

Value between brackets represents break date. NS: non-stationary, S: stationary. *** represents significance at 1%.

Table 6 Break unit root test

Variables	In level	In first difference
LnGRP	– 28.050*** (2021)	– 22.667*** (2021)
LnPOP	– 8.758*** (2004)	– 17.985*** (2011)
LnGFCF	– 26.183*** (2021)	– 24.170*** (2021)
LnEII	– 22.763*** (2021)	– 25.871*** (2021)
LnDFII	– 27.619*** (2004)	– 28.605*** (2021)
LnSO ₂	– 22.874*** (2004)	– 30.567*** (2021)
LnNOX	– 24.434*** (2004)	– 33.162*** (2021)
LnPM2.5	– 22.930*** (2004)	– 34.268*** (2021)

Value between brackets represents break date. *** represents significance at 1%

The cross-sectional dependence (CSD) statistic by Pesaran (2004) is used as the first step after the descriptive analysis to find the correlation coefficients between the time series. The results in Table 3 show the existence of cross-sectional dependence at the 1% level of significance for all variables. Pesaran and Yamagata (2008) developed the slope heterogeneity test, which Swamy (1970) attributed to them. The purpose of the test is to identify the slope heterogeneity or homogeneity between the cross-sections under examination.

The results presented in Table 4 confirm that the null hypothesis of slope homogeneity can be rejected at 1%. Thus, the values for the significant delta tide and the corrected delta tide support the hypothesis that the models' slopes are heterogeneous.

Zhang and Xu (2017), Ganda (2021), Tackie et al. (2020), and Musah et al. (2021) emphasize the importance of conducting a stationary assessment to prevent misleading regression results in panel econometrics. Consequently, this study utilizes the CIPS unit root test introduced by Pesaran (2007), which is considered more robust than first-generation unit root tests. Table 5's findings conclude that the variables integrate at the first difference, represented as $I(1)$.

To verify these results, we move on to examine the stationarity of variables using the Karavias and Tzavalis (2014) break unit root test in Table 6. The detection of significant structural breaks in economic variables provides compelling evidence for utilizing regime-switching models, such as the panel threshold autoregressive (PTAR) and panel smooth transition autoregression (PSTAR) models. These structural breaks reflect changes in the underlying dynamics of variables, indicating the presence of distinct regimes in their behavior over time. As highlighted by Bouattour et al. (2024a, b), regime-switching models are particularly useful in analyzing non-linear relationships where variables transition between different regimes depending on threshold values. By accommodating either abrupt or gradual shifts, PTAR and PSTAR models offer a robust approach for examining complex interactions among economic and financial variables, delivering more precise insights into their influence in varying contexts.

The justification for employing non-linear models becomes even stronger when structural breaks are present in economic data. These breaks signify substantial shifts in variable trajectories, often triggered by significant events or economic adjustments, making linear assumptions insufficient to capture the true dynamics. Non-linear models, such as PTAR and PSTAR, address this limitation by accounting for varying

Table 7 Second-generation and break cointegration tests

Statistic	Value	Z-value	p-value	Robust p-value
Cointegration without rupture (Persyn and Westerlund 2008)				
G_t	− 4.293	− 5.987	0.000	0.000
G_a	− 5.071	3.221	0.999	0.000
P_t	− 18.969	− 8.238	0.000	0.000
P_a	− 6.733	0.247	0.597	0.010
Cointegration with rupture (Ditzen et al. 2021)				
Break	Date	[95% Conf. Interval]		
1	2010	[2009; 2011]		
2	2017	[2015; 2018]		

All test statistics are distributed $N(0,1)$, under a null of no cointegration;

responses across distinct regimes, providing a more realistic and nuanced representation of economic relationships influenced by structural changes. Consequently, these models are essential tools for accurately analyzing the evolving interplay between variables in the presence of structural breaks.

The results in Table 6 confirm the results of Pesaran (2007) results, and consequently, all variables are integrated in the first difference with the existence of some structural breaks on 2004, 2011, and 2021. The identified structural breaks highlight key periods of economic and policy transformation. The 2004 breaks mark early efforts in modernization, environmental regulation, and digital inclusion, driven by globalization, technological adoption, and climate commitments. By 2011, shifts reflected demographic and economic changes, influenced by urbanization, migration, and recovery from the 2008–2009 financial crisis, alongside reforms shaping population dynamics and labor markets. The 2021 breaks underscore post-pandemic recovery, characterized by stimulus-driven growth, accelerated digitalization, green energy transitions, and enhanced environmental policies aimed at resilience, sustainability, and addressing inequalities. These milestones collectively illustrate the evolving interplay between policy and economic development. Thus, it is necessary to test the long-term cointegration relationship between model variables.

The first panel of Table 7 presents the results of second-generation cointegration analysis. The Persyn and Westerlund (2008) test demonstrates that the variables are cointegrated and exhibit a long-term relationship. The results are resilient and precise, offering more accurate critical values in the context of cross-sectional dependency and heterogeneous dynamics, hence affirming a stable long-term equilibrium relationship among the variables.

China's 2010 break, identified by the Ditzen et al. (2023) cointegration test with breaks in the second panel of Table 7, may be related to its post-global financial crisis recovery and sustained development as a global economic power. By 2010, China had recovered from the 2007 to 2008 crisis thanks to its quick fiscal stimulus packages and huge infrastructure spending. This decade also saw China's economic model shift from exports and manufacturing to domestic consumption and services. Trade and FDI were greatly affected by this transition. China's growing influence on global trade policies and supply

chains may explain the 2010 split. Information technology and manufacturing drove new areas of economic growth in China, causing a structural shift in its economy.

China's 2017 break may be linked to numerous economic factors. President Xi Jinping implemented massive economic changes, notably the “made in China 2025” project to modernize China's industrial capability and lessen its dependence on foreign technology. This age prioritized innovation-driven growth in high-tech industries like robots, AI, and clean energy. As the trade war began in 2017, China's export-led economic model faced additional difficulties due to escalating trade tensions, particularly with the US. China's economic links may have been affected by global supply chain disruptions, notably in technology and manufacturing. By 2017, China has also adopted cleaner energy and phased out high-pollution businesses. These developments certainly caused structural changes in China's economy, breaking cointegration ties from earlier years.

In other words, these splits signal major economic shifts in China. The 2010 break may signify the end of China's quick recovery from the global financial crisis and the start of a new economic period focused on domestic consumption and technical innovation. The 2017 split indicates deeper fundamental shifts in China's economy due to technical advances, a shift toward high-tech industries, and new geopolitical issues. These developments may have affected economic processes including exports, manufacturing, and FDI fueling growth. These splits indicate a new economic phase that differs significantly from the period before 2010, emphasizing the necessity of accounting for structural transformations when studying long-term economic linkages in China.

The cross-sectional independence of the residuals in a fixed-effects regression model was evaluated through the application of the Breusch and Pagan (1980) test, which served to substantiate the aforementioned findings. Consequently, the Wald statistic was employed to assess groupwise heteroscedasticity in the residuals of the regression model with fixed effects. This was done on the basis of the Breusch and Pagan (1979) and White (1980) tests. As illustrated in Table 8, the residuals exhibit robust cross-sectional independence, underscoring the prevalence of significant individual heteroscedasticity.

The coefficients are statistically significant (p -value = 0.000) and are associated with a probability of less than 1% (chi2 statistic = 521.4), as indicated by the heteroscedasticity test conclusions. In light of these findings, the homoscedasticity hypothesis (H_0) is rejected. Furthermore, the autocorrelation test indicates that the model demonstrates robust serial autocorrelation, which is statistically significant (p -value = 0.000) and less than 1% (chi2 = 2537.377). In light of the aforementioned evidence, the null hypothesis is thus refuted. These findings corroborate our previous observations regarding the interdependence of individuals and underscore the necessity of incorporating heteroscedasticity into our analytical framework.

Table 8 Autocorrelation and heteroscedasticity tests

Tests	Chi-square statistic	Probability
Wald test groupwise (White 1980) heteroscedasticity	521.40	0.000
Breusch and Pagan (1980) LM cross-sectional independence	2537.377	0.000
Modified Bhargava et al. (1982) Durbin–Watson	0.377	–

Table 9 PTAR linearity test results

Hypotheses	Threshold	F-test	Probability	Confidence interval	RSS	Residual variance
H_0 : No threshold	$c_1 = 3.288$	251.764	0.000	[3.255; 3.298]	1.459	0.0025
H_1 : One threshold	$c_2 = 4.106$			[4.089; 4.359]		

The Durbin–Watson test was applied to panel data from the Bhargava et al. (1982) study to substantiate the presence of a first-order autocorrelation issue. As evidenced in Table 8, the DW statistic is 0.377, which is nearly 0. This corroborates the rejection of the null hypothesis of no first-order autocorrelation in the model.

4.2 PTAR and PSTAR estimations

Non-linear empirical analysis plays a crucial role in contemporary econometric modeling, offering effective tools for exploring the complex interactions between economic variables. In this section, we focus on the application of non-linear models, specifically the PTAR (panel threshold autoregression) and PSTAR (panel smooth transition autoregressive) models. These models provide valuable insights by allowing for a more seamless and progressive estimation process, making them ideal for understanding economic dynamics with thresholds or smooth transitions.

To address the problem of serial autocorrelation in our model, we adopted a p-order PTAR or PSTAR approach (i.e., PTAR (p) or PSTAR (p)). We compared the performance of a zero-lag model with a one-lag model, using the GMM estimator with instrumental variables (GMM-IV). The results showed a Chi-square statistic equal to 1280.946 with a probability of zero. In contrast, when comparing a two-lag model with a one-lag model using the GMM-IV estimator, the Chi-square statistic dropped to 26.282, with a

Table 10 PTAR final results

	LnDFI _{it-1} ≤ 3.288		3.288 < LnDFI _{it-1} ≤ 4.106		LnDFI _{it-1} > 4.106	
	Coefficient	t-statistic	Coefficient	t-statistic	Coefficient	t-statistic
LnGRP _{it-1}	0.780	38.159***	0.645	18.834***	0.791	15.617***
LnPOP _{it}	0.291	4.150***	0.252	4.517***	0.189	3.283***
LnGFCF _{it}	0.035	− 2.678***	0.201	8.421***	0.216	5.458***
LnEll _{it}	− 0.034	− 3.479***	0.025	2.033**	− 0.099	− 5.727***
LnDFI _{it}	0.018	1.651*	0.008	2.370**	0.150	4.183***
LnSO _{2it}	− 0.006	− 0.826	− 0.041	− 2.957***	− 0.022	− 1.957*
LnNOX _{it}	− 0.046	− 2.512**	− 0.016	− 0.688	− 0.043	− 1.948*
LnPM _{2.5it}	− 0.054	− 2.527**	− 0.034	− 2.357**	− 0.049	− 2.972***

***, ** and * represent significance at 1%, 5 and 10%, respectively

Table 11 PSTAR linearity test results

Tests	$H_0: r = 0$ vs $H_1: r = 1$		$H_0: r = 1$ vs $H_1: r = 2$	
	t-Statistic	Probability	t-Statistic	Probability
Wald (LM)	70.565	0.000	15.173	0.056
Fisher (LMF)	9.327	0.000	1.747	0.085
Likelihood ratio (LR)	74.914	0.000	15.361	0.052

Table 12 PSTAR threshold test results

Order	Threshold (c)	Transition parameter (γ)	RSS	AIC	BIC
$m = 1$	$\hat{c}_1 = 2.965$	$\hat{\gamma}_1 = 23.631$	1.465	− 5.910	− 5.709
	$\hat{c}_2 = 4.594$	$\hat{\gamma}_2 = 3.472$			

Table 13 PSTAR final results

	LnDFII _{it-1} ≤ 2.965		2.965 < LnDFII _{it-1} ≤ 4.594		LnDFII _{it-1} > 4.594	
	Coefficient	t-statistic	Coefficient	t-statistic	Coefficient	t-statistic
LnGRP _{it-1}	0.507	6.695***	0.133	4.409***	0.173	2.097**
LnPOP _{it}	0.146	1.846*	− 0.016	− 0.710	0.034	0.706
LnGFCF _{it}	0.376	6.634***	0.143	5.919***	0.200	3.243***
LnEll _{it}	− 0.077	− 2.690***	0.019	2.341**	0.052	2.674***
LnDFII _{it}	0.330	− 4.126***	0.002	2.154**	0.215	2.923***
LnSO _{2it}	0.016	1.322	0.048	3.078***	− 0.052	− 3.033***
LnNOX _{it}	0.057	1.256	− 0.074	− 2.997***	− 0.031	− 2.640***
LnPM _{2.5it}	− 0.149	− 2.347**	− 0.009	− 0.738	− 0.120	− 1.859*

***, **, and * represent significance at 1%, 5%, and 10%, respectively

probability value equal to 1. Based on these comparisons, the PTAR and PSTAR models with one lag ($p = 1$) were identified as the optimal choice for handling the autocorrelation issue. Consequently, we proceed with applying the PTAR model with one lag, i.e., PTAR ($p = 1$). In accordance with Table 9, the 1000-bootstrap procedure showed the existence of two thresholds, LnDFII = 3.288 and LnDFII = 4.106 (i.e., DFII = 26.789 and DFII = 60.703), respectively.

Table 10 displays the PTAR results, indicating that the estimated values are more significant than their signs, which are not easily interpreted. Regardless of their classification as low, intermediate, or high regimes, the results show a positive and statistically significant relationship between digital financial inclusion and regional economic growth across China's provinces. This suggests that the benefits of digital financial inclusion extend across different economic contexts, contributing to growth in various regions, and highlights the importance of enhancing digital financial services as a means to stimulate economic growth.

Testing linearity against the PSTAR serves two purposes: selecting the appropriate transition variable and determining the logistic transition function's order. Table 11 presents overall test results, indicating rejection of the null hypothesis for linear and single-threshold PSTAR models at the 10% significance level.

Therefore, PSTAR with two thresholds presents a compelling alternative, offering a nuanced understanding of the underlying dynamics. In addition, the specification test sequence results indicate the selection of the $m = 1$ transition function. This implies the presence of multiple distinct phases or regimes in the data, each characterized by its own transition function (see Table 12) and there are two thresholds equal to LnDFII = 2.965 and LnDFII = 4.594 (i.e., DFII = 19.395 and DFII = 98.889), respectively.

Fok et al. (2005), González et al. (2005), and González et al. (2017) introduced PSTAR model, facilitating panel observations division into distinct regimes with individual coefficients and enabling the calculation of various elasticities, which can transition smoothly between regimes. Unlike Hansen's (1999) PTAR, the PSTAR model offers advantages by allowing elasticity variations not only over time but also across countries.

The results presented in Table 13 indicate a strong and negative correlation between the digital financial inclusion index (DFII) and gross regional product (GRP). Consequently, as digital financial services become more accessible and widespread, regions experience increased economic activity. In the initial regime, improvements in digital financial inclusion can yield significant economic benefits by expanding financial access and unlocking new economic opportunities that facilitate greater participation in financial markets, enhancing resource allocation, and promoting more inclusive economic growth. However, the intermediate phase reflects a positive but relatively weak effect, where broader access to digital finance results in some improvements in efficiency and cost reductions, but its impact on the economy remains limited.

The strong positive correlation shows that mature digital ecosystems greatly boost economic growth by making it easier for more advanced and efficient financial transactions to happen in the third regime, where digital financial systems are fully developed. The consistent positive relationship across various stages of digital financial inclusion highlights its pivotal role in fostering economic development. Thus, by enhancing access to financial services, reducing transactional costs, and supporting innovation in financial products, digital financial inclusion plays a crucial part in lowering barriers to financial participation and driving productivity across the economy.

The estimated transition functions from the Panel Smooth Transition Auto-Regressive (PSTAR) model elucidate the relationship between digital financial inclusion (DFI) and regional economic growth (GRP) across China's provinces, providing critical insights into how varying levels of financial inclusion influence economic development in different regional contexts.

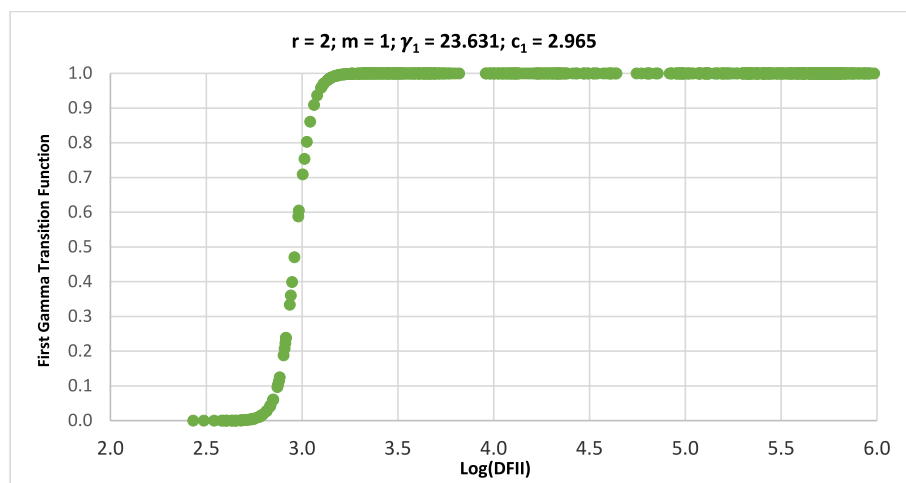


Fig. 2 First transition logistic function of LnDFII on LnGRP

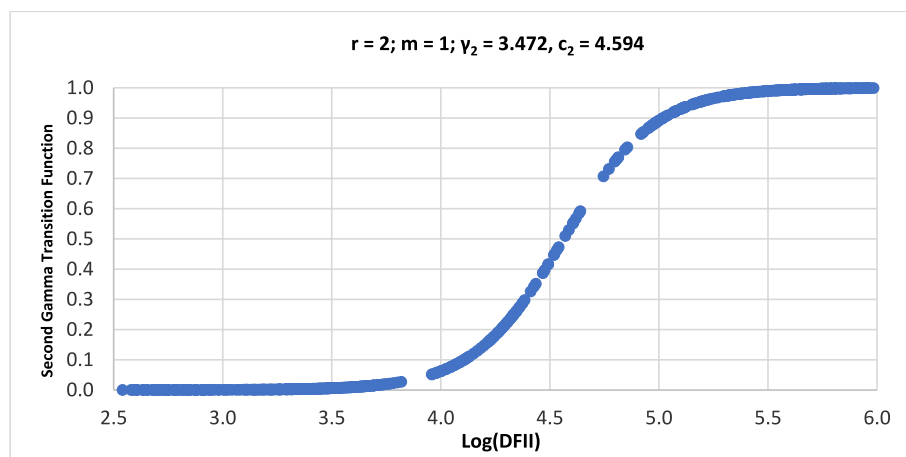


Fig. 3 Second transition logistic function of LnDFII on LnGRP

According to Fig. 2, the transition function shows an estimated transition parameter of $\hat{\gamma}_1 = 23.631$ with a threshold value equal to $\hat{c}_1 = 2.965$. Below this threshold, the impact of digital financial inclusion on economic growth is likely to be less pronounced, suggesting that initial efforts to enhance digital financial services may lead to modest improvements in growth. However, as digital financial inclusion surpasses the threshold, the transition function demonstrates a smooth increase in its influence on economic growth, resulting in accelerated economic development.

Figure 3 illustrates a comparable analysis, showing a value of $\hat{\gamma}_2 = 3.472$ and a threshold of $\hat{c}_2 = 4.594$. The lower value of $\hat{\gamma}$ indicates a more gradual response in economic growth as digital financial inclusion surpasses this threshold. This smooth transition model highlights the complexity and multifaceted nature of the relationship between digital financial inclusion and economic growth.

The results suggest that increases in digital financial inclusion do not lead to immediate or uniform changes in economic growth; instead, the impact unfolds over time and may vary across different contexts or regions. This nuanced understanding is essential for policymakers aiming to harness digital finance for economic development, as it emphasizes the need for tailored strategies that consider the specific circumstances and needs of different stakeholders. By recognizing the intricate dynamics at play, stakeholders can better design interventions that maximize the benefits of digital financial services to promote economic growth.

5 Discussion of the results

The digital financial inclusion expansion has greatly increased financial services accessibility, making them more affordable and effectively reaching rural areas. Moreover, the emergence of low-threshold online lending and money funds has significantly improved living standards in rural areas (Lee et al. 2023). Additionally, the availability of online borrowing options alleviates financial constraints for small and medium-sized enterprises (SMEs), thereby attracting more investment in these regions (Lu et al. 2022).

The establishment of these investment opportunities also leads to job creation, promotes the return of human capital, supports industrial structure upgrading, and ultimately fosters rural economic growth (Yang and Zhang 2020). Additionally, it has facilitated a reallocation of resources, which has contributed to narrowing the urban–rural divide.

Digital financial services offer a variety of opportunities by incorporating technology into a wide array of financial services, promoting inclusive finance through the involvement of financial institutions, credit companies, banks, and similar entities (Zavolokina et al. 2016). These services empower individuals by lowering costs and increasing transparency (Thompson 2017; Anagnostopoulos 2018).

The role of technology in the advancement of inclusive finance is of vital importance not only in China, but also on a global scale (Shim and Shin 2016), which is contingent upon their capacity to mitigate risks and reduce costs within the financial sector in a more efficacious manner relative to the alternatives currently available for financial trading (Frame and White 2012). Moreover, such technologies render inclusive finance more lucrative, prompting banks and other financial institutions to expand their services to low-income customers.

Companies utilizing these technological advantages often achieve greater profitability compared to those that do not. The use of financial technologies helps companies overcome various challenges by improving service quality to meet customer needs (Hao 2017). As a result, even low-income individuals can benefit from technological innovations in the financial sector.

A considerable amount of research has highlighted the positive effects of financial inclusion on human development, as evidenced by the works of Schumpeter (1912), Hicks (1969), Levine (1997), Mohieldin et al. (2019), and Abdelghaffar et al. (2023). Indeed, financial inclusion plays a crucial role in enhancing the Human Development Index (HDI) by improving income, education, life expectancy, and health outcomes. Moreover, it facilitates access to essential financial services, which in turn promotes income growth among economically disadvantaged populations, reduces income inequality, and aids in poverty alleviation (Beck et al. 2007).

In addition, financial inclusion also supports educational attainment, especially for marginalized groups, by providing savings opportunities, thus fostering human capital development (Chand 2018; Anderson et al. 2019). Moreover, education quality, measured by cognitive skills and knowledge, has a stronger correlation with economic growth than quantity and significantly boosts productivity and innovation (Hanushek and Kimko 2000; Barro 2001; Bosworth and Collins 2003; Coulombe and Tremblay 2006).

Due to its very nature, fintech can jeopardize local governments' financial autonomy. This process has numerous forms. FinTech significantly lessens information asymmetry by utilizing big data, the internet, and other cutting-edge technologies (Patel and Sharma 2020). This decrease includes the distance between major state-owned banks' head offices and regional offices as well as the space between banks and regulatory agencies. As a result, this reduces the power that local governments possess. Provincial branches have less influence over lending choices because of the reduced information asymmetry, which strengthens the state-owned bank headquarters' supervisory and decision-making capabilities (Gupta and Singh 2021).

As a result, local governments are less able to control how financial resources are allocated through their dealings with state-owned bank provincial branches. Local governments have historically influenced local banks' operations, including those of urban and rural commercial banks, by interfering with their choices. On the other hand, regulatory authorities obtain far greater authority over local banks when the knowledge gap between them and the banks is reduced, which restricts the capacity of local governments to step in. Second, local governments' power is further diminished by the cost savings made possible by FinTech efforts (Kumar and Mishra 2022).

FinTech reduces transaction costs, especially those associated with acquiring information, and improves operational efficiency by leveraging big data and artificial intelligence (Jin et al. 2020a, b). By increasing bank profitability, these cost-cutting and efficiency gains lessen the reliance of commercial banks, particularly those in rural and urban areas, on local governments, especially when it comes to financing important local projects. As a result, the influence of local governments over financial resources is diminished (Huang and Wu 2020). Thirdly, the autonomy that local governments have historically possessed is diminished by FinTech's use of online operations and intelligent banking. While intelligent banking reduces needless intervention in the financial resource allocation process, online banking effectively distributes financial resources across geographies (Chen and Zhang 2019). By influencing banks, both trends weaken local governments' capacity to direct financial resources.

Digital financial inclusion and the importance of institutional frameworks and governance are critical for fostering sustainable economic growth. China's rapid advancements in digital finance, particularly in digital payments and mobile financial services, have already positioned it as a global leader in fintech. However, for these technological advancements to translate into substantial economic benefits, the Chinese government must prioritize optimizing financial reforms and improving the quality of regulations. One of the most pressing issues for China is optimizing financial reforms to enhance the accessibility of digital financial services.

Further improvements are necessary as China continues to modernize its banking and financial systems to make digital financial services accessible to a larger portion of the population, particularly those in underserved regions like rural areas and low-income groups. These populations often face significant barriers to accessing traditional banking services, and digital financial inclusion can help overcome these obstacles. By removing these barriers, China can enable more people to participate in the economy, boosting productivity and fostering broader economic development.

Given the findings and discussions regarding the influence of digital financial inclusion (DFI) on regional economic growth (GRP) in China, we can support our first hypothesis (H_1), which suggests a non-linear relationship between digital financial inclusion and regional economic growth. This conclusion is backed by the evidence presented, indicating that as digital financial inclusion expands, its impact on regional economic growth varies across different levels of inclusion, highlighting the complexity of this relationship and underscoring the importance of tailored strategies to maximize the benefits of digital financial inclusion in fostering regional economic development.

Furthermore, an educated workforce can adapt more quickly to new technologies and processes, fostering an environment conducive to economic expansion. Additionally,

better health, as a component of human capital, enhances labor force participation and productivity by reducing absenteeism and increasing the overall well-being of the population. Moreover, the findings of Oketch (2006) and Ogundari and Awokuse (2018) underscore the importance of policies that promote access to quality education and healthcare. By investing in human capital, governments can lay the foundation for economic growth, as a more educated and healthier workforce is capable of driving innovation, increasing productivity, and ultimately boosting the economy's output.

Domestic investment exhibits positive and statistically significant coefficients, underscoring its pivotal role in promoting economic growth, which posits that increased domestic investment enhances productive capacity, stimulates employment, and fosters technological advancement, all of which contribute to economic expansion. For instance, Adams (2009) emphasizes that domestic investment serves as a key driver of economic growth by facilitating capital accumulation and enabling the efficient allocation of resources. Similarly, Eregha (2012) underscores that the positive impact of domestic investment on growth is particularly pronounced in developing economies, where it acts as a catalyst for industrialization and infrastructure development.

Foreign direct investment (FDI) plays a crucial role in increasing China's capital stock and fostering high-quality economic development. FDI can boost capital formation through large-scale investments and, if local supply is insufficient, it encourages additional investments from multinational corporations (MNCs) via parent firms or partners, thereby indirectly enhancing capital stock. Furthermore, FDI can mobilize local savings by offering attractive investment opportunities, acting as a catalyst for local investments. When FDI enters as greenfield investment, it directly increases both savings and investment, significantly raising capital stock. Alternatively, FDI through mergers and acquisitions may not immediately boost capital stock but enhances production capacity, benefiting capital in the long term. Additionally, FDI indirectly promotes high-quality economic development by fostering innovation. According to monopoly advantage theory, MNCs engage in direct investment in China due to its large market and competitive advantages. MNCs accelerate the construction of factories and R&D centers, leading to localized innovation. While China may not easily acquire cutting-edge technology, FDI entry prompts changes in the intellectual property system, capital investment, management, and competitive market structure, all of which drive innovation and contribute to economic growth (Zhang et al. 2024).

Energy industrial investments have a significant and multifaceted impact on regional economic growth in China, particularly by spurring infrastructure development, generating employment, and driving technological progress. Investments in renewable energy sectors such as wind, solar, and hydropower have been especially transformative, providing substantial benefits to local economies. The broader economic effects of energy industrial investments extend beyond immediate employment gains. By improving energy infrastructure, particularly through the promotion of clean technologies, these investments lead to enhanced productivity and long-term economic resilience.

Local firms gain the opportunity to innovate and adopt more energy-efficient practices, which improves their competitiveness and drives technological advancements. Furthermore, these investments attract both domestic and foreign capital, further

boosting local economies and fostering the growth of a diverse industrial base. The broader economic impact includes an increase in the capacity for innovation, leading to more efficient production methods and, ultimately, greater economic diversification and sustainability. However, energy investments distribution across China has been uneven, contributing to significant regional disparities in economic development. Coastal provinces attract the majority of energy investments due to their advanced infrastructure and stronger economic positions, led to rapid industrialization and technological advancement in these regions, strengthening their economies. On the other hand, inland provinces, which often face infrastructure challenges and have a less developed industrial base, struggle to attract comparable investments.

The relationship between technological innovation and sustainability is also critical. Technological advancements, especially those related to eco-friendly appliances and waste management systems, can drive sustainable practices within industries. In China, where pollution and environmental degradation have become pressing issues, these green technologies represent essential strategies for environmental preservation (Dimnwobi et al. 2023). Financial inclusion supports the diffusion of these technologies by making them accessible to a larger portion of the population. As these technologies gain wider acceptance, they promote ecological sustainability and shift societal norms toward more environmentally conscious behaviors (Dimnwobi et al. 2021). The diffusion of these green practices, supported by FI, could create a snowball effect that accelerates the transition toward a more sustainable economy, as seen in the growing adoption of renewable energy technologies (Onuoha et al. 2023).

Digital financial inclusion integrates traditional financial inclusion with digital technologies. Sarma and Pais (2011) demonstrate a strong positive correlation between financial inclusion and economic development. Moreover, enhancing financial inclusion can reduce the barriers to achieving economic sustainability (Sethi and Acharya 2018). The financial environment plays a crucial role in influencing the effect of environmental regulations on economic development. Implementing these regulations involves long-term, high-cost investments, which place significant financial strain (Chen et al. 2022). However, digital finance, with its low cost, broad accessibility, and sustainability, helps to relieve this pressure, making it an essential factor in the link between environmental regulation and sustainable economic growth (Yang et al. 2022).

China has capitalized on the internet revolution to develop a comparative advantage in digital finance. This has reduced the barriers of “difficult and expensive financing”, enabling businesses to invest in clean technology research and development, enhancing both innovation and output (Drasch et al. 2018; Liu and Chen 2022). Additionally, digital finance supports industrial upgrading by favoring service sectors, which benefit from data-driven credit decisions and contribute more to environmental sustainability compared to traditional finance, which typically favors manufacturing due to collateral-based credit systems (Wan et al. 2020).

Furthermore, digital finance mitigates the negative environmental impact of energy-intensive manufacturing and encourages cleaner practices in agriculture. By reducing information asymmetry, it ensures a stable supply of credit to farming communities, promoting sustainable agriculture (Anshari et al. 2019; Yu et al. 2020; Li et al. 2021).

Green finance plays a vital role in promoting environmental protection and governance by redirecting resources from high-pollution, energy-intensive industries to sectors driven by advanced technologies and sustainable practices. Power finance, as a subset of green finance, complements this effort, while digital finance, a cornerstone of the digital economy, represents a key innovation pathway for advancing green finance and financial technology. To accelerate the integration of digital technology with financial development, it is essential to align the long-term benefits of investment projects with ecological goals, prioritize the growth of green industries, and support sustainable social and economic development.

Under the global vision of “carbon neutrality”, the shift to a low-carbon economy has become a shared priority, with “green finance” and “digital finance” serving as critical tools to balance industrial growth and ecological preservation. First, trends in electric financing significantly enhance energy efficiency, while improved regional innovation capabilities bolster the role of digital finance in achieving higher energy efficiency. Second, both digital finance and regional innovation influence energy utilization efficiency differently across regions and generate positive spatial spillover effects, enhancing energy efficiency in neighboring areas. Thus, it is imperative to develop robust digital infrastructure, such as 5G, artificial intelligence, and big data, to support the evolution of digital financial services. Third, accelerating enterprise-led innovation is crucial for improving energy efficiency, reducing environmental pollution, and enhancing China’s technological innovation capacity through digital finance. This synergy fosters advancements in financial accounting and promotes cooperative, win–win interactions between digital and green finance (Yang and Masron, 2022).

From a policy perspective, first, the government increases financing and subsidies for clean energy production and processing enterprises, which can reduce the cost of clean energy use and increase the choice of energy consumption for various economic entities. Therefore, the government will gradually replace traditional high-carbon energy sources (Fareed et al. 2022). On the other hand, the government imposes financing constraints and pollution control measures on polluting enterprises to encourage them to fulfill their environmental and social responsibilities (Konadu et al. 2022; Lv and Bai 2021). Therefore, enterprises can reduce carbon emissions by improving green technology innovation capabilities (Tang et al. 2020).

Financial institutions, as an essential hub for optimizing resource allocation, are responsible for providing a favorable investment and financing environment for the construction of a green and low-carbon new energy supply and demand system (Shahbaz et al. 2022). The participation of financial institutions at the macro-level in climate-related investment and financing projects can reduce the country’s total carbon emissions (Yang et al. 2023a, b).

Local-level financial institutions can rely on their financing and investment capabilities, based on policy guidance and professional services, to help regional low-carbon work and improve ecological and economic benefits (Zheng and Li 2022). From the perspective of industrial structure, DFI can promote technological progress and the upgrading of industrial structure (Cheng et al. 2023). Further, it can promote the development of the service and high-tech industries and improve the quality of industrial integration (Wu et al. 2023), thereby reducing carbon emissions.

From the perspective of enterprises, DFI can impose credit constraints on high-carbon emission enterprises (Yang and Zhang 2020) and force enterprises to improve green investment efficiency (He et al. 2022) and green technology innovation level (Xiong et al. 2022). For enterprises with significant carbon reduction benefits, DFI can prioritize using more convenient financial services and more favorable financing costs to stimulate their motivation to save energy and reduce emissions (Xue and Zhang 2022). From the perspective of consumers, DFI increases the disposable income of residents through inclusive means (Xiong et al. 2022). At the same time, relying on information-based payment means increasing residents' demand for green consumption (Zhang et al. 2023a, b), which in turn reduces emissions.

Economic activity, environmental policy, and the integration of digital finance have all played a role in the substantial changes in the evolution of pollutants—sulfur dioxide (SO_2), nitrogen oxides (NO_x), and particulate matter ($\text{PM}_{2.5}$)—across Chinese provinces during the post-COVID-19 period. The early stages of the pandemic were characterized by a significant decrease in pollutant levels, which was notably apparent in SO_2 and NO_x emissions, as a result of the widespread lockdowns and reductions in industrial output. Furthermore, vehicle traffic and industrial production experienced a significant decline. Nevertheless, emissions resumed to increase in late 2020 and 2021, particularly in regions that have a high reliance on traditional, energy-intensive industries. This reflects the difficulties of achieving long-term reductions when economic recovery prioritizes industrial output. On the other hand, the pandemic also facilitated a transition to green recovery strategies, as the Chinese government prioritized environmental sustainability and cleaner technologies as part of its recovery initiatives.

The transition to a green economy has been more pronounced in regions with higher levels of digital financial inclusion. The utilization of digital platforms in these regions has effectively promoted green finance, resulting in significant reductions in pollutants and investments in energy-efficient technologies and renewable energy. Furthermore, digital finance has been instrumental in the allocation of resources to sustainable sectors, thereby reducing the environmental impact of industrial activities. Nevertheless, despite the successful integration of digital finance and green technologies to reduce pollution in more economically advanced regions, less developed areas continue to encounter significant obstacles in adoption of cleaner energy solutions. The same environmental improvements that are observed in the more digitally advanced regions have been difficult to achieve in these regions, which are still reliant on coal and other high-polluting industries.

The preceding discussions suggest that digital financial inclusion positively influences environmental sustainability, particularly concerning the reduction of air pollution. This assertion is supported by the widespread implementation of digital financial inclusion (DFI) in Regime 3. Consequently, we can affirm the validity of hypothesis $\mathbf{H}_2$, which posits that digital financial inclusion contributes to enhancing environmental outcomes. The evidence highlights the crucial role that digital financial services play in promoting sustainable practices, thereby reinforcing the connection between financial inclusion and environmental benefits.

Thus, the two hypotheses ($\mathbf{H}_1$ and $\mathbf{H}_2$) outlined above have been validated through the findings of our study. The evidence supports the notion that there is a significant

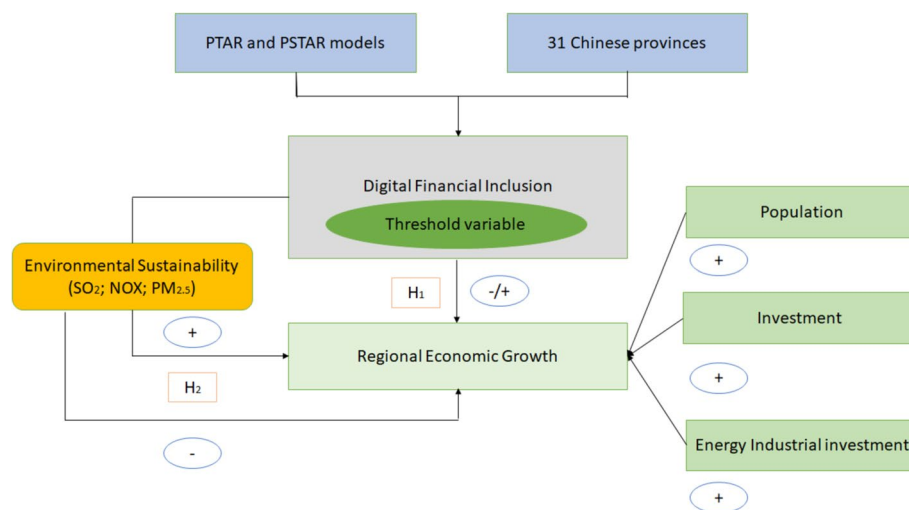


Fig. 4 Overview of results and validation of hypotheses

non-linear relationship between digital financial inclusion and economic growth. Additionally, our study confirms the hypothesis that digital financial inclusion positively impacts environmental sustainability by reducing air pollution. The results indicate that increased access to digital financial services contributes to reducing air pollution and promoting the adoption of cleaner technologies, which underscores the importance of recognizing the dual benefits of digital financial inclusion in enhancing economic growth and fostering environmental sustainability.

In Fig. 4, we present an overview of the approaches used, the sample, and the main results found with the validation of hypotheses.

6 Conclusion and policy implications

In the context of the Chinese economy, digital financial inclusion and environmental pollution are key factors influencing regional growth dynamics. Digital financial inclusion, by facilitating access to financial services through digital technologies, stimulates investment, innovation, and economic efficiency. Conversely, environmental pollution, particularly the presence of pollutants such as SO_2 , NOX , and $\text{PM}_{2.5}$, can hinder this growth by impacting public health, productivity, and the attractiveness of investments. Therefore, the interaction between these two dimensions becomes crucial to understanding how Chinese provinces can achieve sustainable economic growth while minimizing negative environmental impacts.

In this context, this study aims to investigate the non-linear effect between digital inclusion and regional economic growth for 31 Chinese provinces using the panel threshold autoregressive model (PTAR) and the panel smooth transition autoregressive model (PSTAR). The results prove that there exists a non-linear relationship between digital financial inclusion and regional economic growth with two thresholds equal to $\text{LnDFII}=3.288$ and $\text{LnDFII}=4.106$ (i.e., $\text{DFII}=26.789$ and $\text{DFII}=60.703$) for the PTAR model, and $\text{LnDFII}=2.965$ and $\text{LnDFII}=4.594$ (i.e., $\text{DFII}=19.395$ and $\text{DFII}=98.889$) for the PSTAR model, respectively. In addition, the decrease of pollutants (SO_2 , NOX , and $\text{PM}_{2.5}$) can support the regional economic growth of Chinese

provinces through the inclusion of digital financial services that help to invest in clean energy and consequently contribute to more sustainable economic growth.

The findings of this study offer several practical implications for policymakers, businesses, and financial institutions aiming to foster sustainable growth in China. First, the non-linear relationship between digital financial inclusion (DFI) and gross regional product (GRP) suggests differentiating policies based on the level of digital financial infrastructure in different provinces. Regions with lower levels of DFI could prioritize investments in digital infrastructure, such as expanding mobile banking services and internet connectivity, to ensure that underserved populations gain access to financial services. In contrast, more digitally advanced regions could focus on enhancing the quality of financial services by introducing advanced technologies like blockchain, AI, and big data analytics to increase efficiency and drive innovation. By tailoring policies to the unique needs of each province, China can create a more inclusive and sustainable digital financial ecosystem that stimulates regional economic growth.

In addition, the study highlights the importance of integrating environmental considerations into the financial inclusion agenda. Encouraging investment in clean energy technologies and promoting sustainable practices through financial incentives—such as subsidies for renewable energy projects or tax breaks for green businesses—could help mitigate the adverse effects of environmental pollution. Policymakers should encourage businesses to adopt green technologies by leveraging digital financial services, thus ensuring that economic growth does not come at the expense of the environment. These policy interventions can also drive long-term benefits, such as improved public health outcomes, enhanced productivity, and increased attractiveness of regions for foreign investments, contributing to a more resilient and sustainable economy.

Furthermore, fostering collaboration between provinces is crucial for sharing knowledge, best practices, and resources that could accelerate both digital financial inclusion and environmental sustainability efforts. Inter-regional partnerships could be particularly beneficial in harmonizing standards for digital financial services, improving regional environmental policies, and promoting coordinated efforts to tackle pollution. A unified approach to digital inclusion and green investments would ensure that all regions benefit from economic growth and environmental protection, reducing disparities between more developed and less developed areas. This collaborative effort could also provide a platform for Chinese provinces to engage with international stakeholders, further advancing China's position as a leader in economic development.

As with any research, there are some limitations in the study. Firstly, it focuses on the role of financial inclusion and pollution reduction, but it does not explore other potential mediating factors, such as education, infrastructure quality, or government policies, which could also influence growth dynamics. Future research could incorporate a broader range of variables and consider potential interactions between these factors to create a more comprehensive model of regional economic development.

Secondly, it focuses solely on China, which, while providing rich insights into the country's economic and environmental context, may limit the generalizability of the findings to other regions with different economic structures, digital financial landscapes,

and environmental challenges. Comparative studies across other emerging economies could help test the applicability of these findings in different settings and provide insights into how digital financial inclusion and environmental sustainability intersect in other parts of the world. This would offer valuable lessons for other nations striving to achieve sustainable economic growth while addressing environmental concerns.

Author contributions

All authors contributed to the methodology and writing of the different sections of the article. Professor Kamel HELALI and Professor Maha KALAI carried out the empirical part based on the available data. Doctor Hamdi BECHA and Ph.D. Student Saïfeddine HOUIDI wrote and revised the document. Professor Maha KALAI supervised the work.

Funding

This project was completed without any funding.

Availability of data and materials

The China National Bank of Statistics provided the relative variables for our panel, which represents 31 Chinese provinces between 2003 and 2022. The proposed framework comprises several key variables, including gross regional product (GRP), which represents an economic measure equivalent to gross domestic product (GDP) but, at a regional level, quantifies the total economic output produced within a province in a given period. We use Stata 17.0 and Matlab R21 software to run different programs. The data are available on request from the corresponding author.

Declarations

Competing interests

We declare this manuscript to be original, unpublished, and not currently under consideration for publication elsewhere. We confirm that the manuscript has been read and approved by all named authors, and that there are no other people who meet the criteria for authorship but are not listed. We further affirm that each of us has approved the order of authors listed in the manuscript. We understand that the corresponding author is the sole contact for the editorial process. We are responsible for communicating with the other authors about progress, submissions of revisions, and final approval of proofs. The authors are responsible for the correctness of the statements provided in the manuscript.

Received: 30 September 2024 Accepted: 11 January 2025

Published online: 12 February 2025

References

- Abdallah AB, Becha H, Kalai M, Helali K (2024) Development of digital financial inclusion in China's regional economy: evidence from panel threshold models. *J Telecommun Dig Econ* 12(1):637–656. <https://doi.org/10.18080/jtde.v12n1.838>
- Abdelghaffar RA, Emam HA, Samak NA (2023) Financial inclusion and human development: is there a nexus? *J Hum Appl Soc Sci* 5(3):163–177. <https://doi.org/10.1108/JHASS-11-2021-0178>
- Adams S (2009) Foreign direct investment, domestic investment, and economic growth in Sub-Saharan Africa. *J Policy Modeling* 31(6):939–949. <https://doi.org/10.1016/j.jpolmod.2009.03.003>
- Anagnostopoulos I (2018) Fintech and regtech: impact on regulators and banks. *J Econ Bus* 100:7–25. <https://doi.org/10.1016/j.jeconbus.2018.07.003>
- Anderson S Hopkins D, Valenzuela M (2019) The role of financial services in youth education and employment. Working paper. CGAP, Washington, DC. Available online: http://www.ruralfinanceandinvestment.org/sites/default/files/2019_06_30_WorkingPaper_Youth_Education_Employment.pdf (accessed on 12 January 2021)
- Anshari M, Almunawar MN, Masri M, Hamdan M (2019) Digital marketplace and FinTech to support agriculture sustainability. *Energy Procedia* 156:234–238. <https://doi.org/10.1016/j.egypro.2018.11.134>
- Arjunwadkar PY (2018) *FinTech: the technology driving disruption in the financial services industry*. Auerbach Publications
- Aziz A, Naima U (2021) Rethinking digital financial inclusion: evidence from Bangladesh. *Technol Soc* 64:101509. <https://doi.org/10.1016/j.techsoc.2020.101509>
- Bakhsh K, Akmal T, Ahmad T, Abbas Q (2022) Investigating the nexus among sulfur dioxide emission, energy consumption, and economic growth: empirical evidence from Pakistan. *Environ Sci Pollut Res* 29(5):7214–7224. <https://doi.org/10.1007/s11356-021-15898-9>
- Bakhsh S, Zhang W, Ali K, Anas M (2024) Transition towards environmental sustainability through financial inclusion, and digitalization in China: evidence from novel quantile-on-quantile regression and wavelet coherence approach. *Technol Forecast Soc Chang* 198:123013. <https://doi.org/10.1016/j.techfore.2023.123013>
- Banna H, Alam MR (2021) Impact of digital financial inclusion on ASEAN banking stability: implications for the post-Covid-19 era. *Stud Econ Financ* 38(2):504–523. <https://doi.org/10.1108/SEF-09-2020-0388>
- Barro RJ (2001) Human capital and growth. *Am Econ Rev* 91(2):12–17. <https://doi.org/10.1257/aer.91.2.12>
- Basnayake D, Naranpanawa A, Selvanathan S, Bandara JS (2024) Financial inclusion through digitalization and economic growth in Asia-Pacific countries. *Int Rev Financ Anal* 96:103596. <https://doi.org/10.1016/j.irfa.2024.103596>

- Beck T, Demirgüç-Kunt A, Levine R (2007) Finance, inequality and the poor. *J Econ Growth* 12(1):27–49. <https://doi.org/10.1007/s10887-007-9010-6>
- Beck T, Pamuk H, Ramrattan R, Uras BR (2018) Payment instruments, finance and development. *J Dev Econ* 133:162–186. <https://doi.org/10.1016/j.jdeveco.2018.01.005>
- Bhargava A, Franzini L, Narendranathan W (1982) Serial correlation and the fixed effects model. *Rev Econ Stud* 49(4):533–549. <https://doi.org/10.2307/2297285>
- Born B, Breitung J (2016) Testing for serial correlation in fixed-effects panel data models. *Econom Rev* 35(7):1290–1316. <https://doi.org/10.1080/07474938.2014.976524>
- Bosworth BP, Collins SM (2003) The empirics of growth: an update. *Brookings Papers Econ Activity* 2003(2):113–179
- Bouattour A, Kalai M, Helali K (2023) The nonlinear impact of technologies import on industrial employment: a panel threshold regression approach. *Heliyon*. <https://doi.org/10.1016/j.heliyon.2023.e20266>
- Bouattour A, Kalai M, Helali K (2024a) The non-linear relationship between ESG performance and bank stability in the digital era: new evidence from a regime-switching approach. *Hum Soc Sci Commun* 11(1):1–17. <https://doi.org/10.1057/s41599-024-03876-8>
- Bouattour A, Kalai M, Helali K (2024b) Threshold effects of technology import on industrial employment: a panel smooth transition regression approach. *J Econ Struct* 13(1):1–33. <https://doi.org/10.1186/s40008-023-00322-x>
- Breusch TS, Pagan AR (1979) A simple test for heteroscedasticity and random coefficient variation. *Econometrica*. <https://doi.org/10.2307/1911963>
- Breusch TS, Pagan AR (1980) The Lagrange multiplier test and its applications to model specification in econometrics. *Rev Econ Stud* 47(1):239–253. <https://doi.org/10.2307/2297111>
- Bruhn M, Love I (2014) The real impact of improved access to finance: evidence from Mexico. *J Financ* 69(3):1347–1376. <https://doi.org/10.1111/jofi.12091>
- Cao S, Nie L, Sun H, Sun W, Taghizadeh-Hesary F (2021) Digital finance, green technological innovation and energy-environmental performance: evidence from China's regional economies. *J Clean Prod* 327:129458. <https://doi.org/10.1016/j.jclepro.2021.129458>
- Chand M (2018) Aging in South Asia: challenges and opportunities. *South Asian J Bus Stud* 7(2):189–206. <https://doi.org/10.1108/SAJBS-09-2017-0103>
- Chang L, Iqbal S, Chen H (2023) Does financial inclusion index and energy performance index co-move? *Energy Policy* 174:113422. <https://doi.org/10.1016/j.enpol.2023.113422>
- Chaudhry IS, Yusop Z, Habibullah MS (2022) Financial inclusion–environmental degradation nexus in OIC countries: new evidence from environmental Kuznets curve using DCCE approach. *Environ Sci Pollut Res* 29(4):5360–5377. <https://doi.org/10.1007/s11356-021-15941-9>
- Chen L, Zhang H (2019) Financial technology and economic growth: evidence from India. *J Financ Res* 10(4):567–582
- Chen J, Cui H, Xu Y, Ge Q (2021) Long-term temperature and sea-level rise stabilization before and beyond 2100: estimating the additional climate mitigation contribution from China's recent 2060 carbon neutrality pledge. *Environ Res Lett* 16(7):074032. <https://doi.org/10.1088/1748-9326/ac0cac>
- Chen H, Qi S, Zhang J (2022) Towards carbon neutrality with Chinese characteristics: from an integrated perspective of economic growth–equity–environment. *Appl Energy* 324:119719. <https://doi.org/10.1016/j.apenergy.2022.119719>
- Cheng X, Yao D, Qian Y, Wang B, Zhang D (2023) How does fintech influence carbon emissions: evidence from China's prefecture-level cities. *Int Rev Financ Anal*. <https://doi.org/10.1016/j.irfa.2023.102655>
- Cobb CW, Douglas PH (1928) A theory of production. *Am Econ Rev* 18(1):139–165. <https://www.jstor.org/stable/1811556>
- Cole MA, Elliott RJ (2005) FDI and the capital intensity of “dirty” sectors: a missing piece of the pollution haven puzzle. *Rev Dev Econ* 9(4):530–548. <https://doi.org/10.1111/j.1467-9361.2005.00292.x>
- Coulombe S, Tremblay JF (2006) Migration, human capital, and skills redistribution across the Canadian provinces. *Industry Canada, Micro-Economic Policy and Analysis*
- Demirguc-Kunt A, Klapper L, Singer D (2017) Financial Inclusion and Inclusive Growth: A Review of Recent Empirical Evidence. *World Bank Policy Research Working Paper No. 8040*. <https://ssrn.com/abstract=2958542>
- Dimnwoi SK, Ekesiobi C, Madichie CV, Asongu SA (2021) Population dynamics and environmental quality in Africa. *Sci Total Environ* 797:149172. <https://doi.org/10.1016/j.scitotenv.2021.149172>
- Dimnwoi SK, Okere KI, Onuoha FC, Uzoechina BI, Ekesiobi C, Nwokoye ES (2023) Energizing environmental sustainability in Sub-Saharan Africa: the role of governance quality in mitigating the environmental impact of energy poverty. *Environ Sci Pollut Res* 30(45):101761–101781. <https://doi.org/10.1007/s11356-023-29541-2>
- Dinda S (2004) Environmental Kuznets curve hypothesis: a survey. *Ecol Econ* 49(4):431–455. <https://doi.org/10.1016/j.ecolecon.2004.02.011>
- Ding R, Shi F, Hao S (2022) Digital inclusive finance, environmental regulation, and regional economic growth: an empirical study based on spatial spillover effect and panel threshold effect. *Sustainability* 14(7):4340. <https://doi.org/10.3390/su14074340>
- Ditzen J, Karavias Y, Westerlund J (2021) Xtbreak: estimating and testing for structural breaks in stata. In: *US Stata Conference 2021*. University of Bolzano, Italy. August 5, 2021. Retrieved from: http://fmwww.bc.edu/RePEc/scon2021/US21_Ditzen.pdf. (Accessed 15 December 2022)
- Drasch BJ, Schweizer A, Urbach N (2018) Integrating the ‘Troublemakers’: a taxonomy for cooperation between banks and fintechs. *J Econ Bus* 100:26–42. <https://doi.org/10.1016/j.jeconbus.2018.04.002>
- Duvendack M, Mader P (2020) Impact of financial inclusion in low-and middle-income countries: a systematic review of reviews. *J Econ Surveys* 34(3):594–629. <https://doi.org/10.1111/joes.12367>
- Eregba PB (2012) The dynamic linkages between foreign direct investment and domestic investment in ECOWAS countries: a panel cointegration analysis. *Afr Dev Rev* 24(3):208–220. <https://doi.org/10.1111/j.1467-8268.2012.00317.x>
- Fareed Z, Rehman MA, Adebayo TS, Wang Y, Ahmad M, Shahzad F (2022) Financial inclusion and the environmental deterioration in Eurozone: the moderating role of innovation activity. *Technol Soc* 69:101961. <https://doi.org/10.1016/j.techsoc.2022.101961>

- Feng S, Chong Y, Yu H, Ye X, Li G (2022) Digital financial development and ecological footprint: evidence from green-biased technology innovation and environmental inclusion. *J Clean Prod* 380:135069. <https://doi.org/10.1016/j.jclepro.2022.135069>
- Fok D, Van Dijk D, Franses PH (2005) A multi-level panel STAR model for US manufacturing sectors. *J Appl Econom* 20(6):811–827. <https://doi.org/10.1002/jae.822>
- Frame WS, White LJ (2012) Oxford handbooks online technological change, financial innovation, and diffusion in banking, 1–25. Retrieved from <https://doi.org/10.1093/oxfordhb/9780199640935.013.0019>
- Gabor D, Brooks S (2017) The digital revolution in financial inclusion: international development in the fintech era. *New Polit Econ* 22(4):423–436. <https://doi.org/10.1080/13563467.2017.1259298>
- Ganda F (2021) The influence of growth determinants on environmental quality in Sub-Saharan Africa states. *Environ Dev Sustain* 23(5):7117–7139. <https://doi.org/10.1007/s10668-020-00907-7>
- González A, Teräsvirta T, Dijk DV (2005) Panel smooth transition regression models (No. 604). SSE/EFI Working Paper Series in Economics and Finance. <https://hdl.handle.net/10419/56363>
- González A, Teräsvirta T, van Dijk D, Yang Y (2017) Panel Smooth Transition Regression Models. SSE/EFI Working Paper Series in Economics and Finance 604, Stockholm School of Economics, revised 11 Oct 2017. <https://EconPapers.repec.org/RePEc:aah:create:2017-36>
- Gu B, Chen F, Zhang K (2021) The policy effect of green finance in promoting industrial transformation and upgrading efficiency in China: analysis from the perspective of government regulation and public environmental demands. *Environ Sci Pollut Res* 28(34):47474–47491. <https://doi.org/10.1007/s11356-021-13944-0>
- Gupta M, Singh P (2021) Empirical analysis of FinTech impact on India's digital economy. *J Financ Res* 8(2):345–362
- Han C, Gu Z, Yang H (2021) EKC test of the relationship between nitrogen dioxide pollution and economic growth—a spatial econometric analysis based on Chinese city data. *Int J Environ Res Public Health* 18(18):9697. <https://doi.org/10.3390/ijerph18189697>
- Hansen BE (1999) Threshold effects in non-dynamic panels: estimation, testing, and inference. *J Econom* 93(2):345–368. [https://doi.org/10.1016/S0304-4076\(99\)00025-1](https://doi.org/10.1016/S0304-4076(99)00025-1)
- Hanushek EA, Kimko DD (2000) Schooling, labor-force quality, and the growth of nations. *Am Econ Rev* 90(5):1184–1208. <https://doi.org/10.1257/aer.90.5.1184>
- Hao CJ (2017) A study of inclusive financial system in China. *Res J Finance Account* 8(24):27–35
- Hao Y, Liu YM (2016) The influential factors of urban PM2.5 concentrations in China: a spatial econometric analysis. *J Clean Prod* 112:1443–1453. <https://doi.org/10.1016/j.jclepro.2015.05.005>
- He G, Shen L (2021) Whether digital financial inclusion can improve capital misallocation or not: a study based on the moderating effect of economic policy uncertainty. *Discret Dyn Nat Soc* 2021(1):4912836. <https://doi.org/10.1155/2021/4912836>
- He Z, Chen H, Hu J, Zhang Y (2022) The impact of digital inclusive finance on provincial green development efficiency: empirical evidence from China. *Environ Sci Pollut Res* 29(60):90404–90418. <https://doi.org/10.1007/s11356-022-22071-3>
- Helali K, Kalai M (2021) Threshold effect of foreign direct investment on economic growth: new evidence from a panel regime switching models. *Global Bus Econ Rev* 25(2):177–205. <https://doi.org/10.1504/GBER.2021.118212>
- Hicks J (1969) A theory of economic history. Clarendon Press, Oxford
- Hu Y, Dai X, Zhao L (2022) Digital finance, environmental regulation, and green technology innovation: an Empirical Study of 278 Cities in China. *Sustainability* 14(14):8652. <https://doi.org/10.3390/su14148652>
- Huang Y, Wang X (2011) Does financial repression inhibit or facilitate economic growth? A case study of Chinese reform experience. *Oxford Bull Econ Stat* 73(6):833–855. <https://doi.org/10.1111/j.1468-0084.2011.00677.x>
- Huang Q, Wu J (2020) Impact of digital economy on economic development: a comprehensive review. *J Econ Growth* 15(3):321–335
- Jalil A, Feridun M (2011) The impact of growth, energy and financial development on the environment in China: a cointegration analysis. *Energy Econ* 33(2):284–291. <https://doi.org/10.1016/j.eneco.2010.10.003>
- Jarque CM, Bera AK (1987) A test for normality of observations and regression residuals. *Int Stat Rev/revue Internationale De Statistique* 55(2):163–172. <https://doi.org/10.2307/1403192>
- Jiang H, Elahi E, Gao M, Huang Y, Liu X (2024) Digital economy to encourage sustainable consumption and reduce carbon emissions. *J Clean Prod* 443:140867. <https://doi.org/10.1016/j.jclepro.2024.140867>
- Jin X, Li C, Wang Y (2020a) The implications of new technology in the financial industry. *J Financ Innov* 6(2):45–59
- Jin X, Yan C, Xu Y, Huang Z (2020b) The impact of Fintech on financial system: a bibliometric analysis and literature review. *Technol Forecast Soc Chang* 161:120256
- Kalai M, Becha H, Helali K (2024a) Threshold effect of foreign direct investment on economic growth in BRICS countries: new evidence from PTAR and PSTAR models. *Int J Econ Policy Stud* 18(1):227–258. <https://doi.org/10.1007/s42495-023-00126-8>
- Kalai M, Becha H, Kamel H (2024b) Exploring the nexus link of foreign direct investment inflows and openness on economic growth: evidence from BRICS economies. *J East-West Bus*. <https://doi.org/10.1080/10669868.2024.2372768>
- Karavias Y, Tzavalis E (2014) Testing for unit roots in short panels allowing for a structural break. *Comput Stat Data Anal* 76:391–407. <https://doi.org/10.1016/j.csda.2012.10.014>
- Konadu R, Ahinful GS, Boakye DJ, Elbardan H (2022) Board gender diversity, environmental innovation and corporate carbon emissions. *Technol Forecast Soc Chang* 174:121279. <https://doi.org/10.1016/j.techfore.2021.121279>
- Kumar V, Mishra S (2022) Role of local financial regulatory mechanisms in India's digital economy. *J Econ Policy* 18(4):221–236
- Lai JT, Yan IK, Yi X, Zhang H (2020) Digital financial inclusion and consumption smoothing in China. *Chin World Econ* 28(1):64–93. <https://doi.org/10.1111/cwe.12312>
- Lai JT, Xie J, Cao S, Zhang H (2022) Digital financial inclusion and illegal fundraising in China. *Appl Econ* 54(48):5575–5590. <https://doi.org/10.1080/00036846.2022.2047601>
- Lardy N (2008) Financial repression in China. Peterson Institute for International Economics Working Paper No. PB08-8

- Lee CC, Lou R, Wang F (2023) Digital financial inclusion and poverty alleviation: evidence from the sustainable development of China. *Econ Anal Policy* 77:418–434. <https://doi.org/10.1016/j.eap.2022.12.004>
- Levine R (1997) Financial development and economic growth: views and agenda. *J Econ Lit* 35(2):688–726
- Levine R (2005) Finance and growth: theory and evidence. *Handbook Econ Growth* 1(A):865–934. [https://doi.org/10.1016/S1574-0684\(05\)01012-9](https://doi.org/10.1016/S1574-0684(05)01012-9)
- Li J (2024) Examining the impact of digital financial inclusion on economic development in urban and rural areas of China using remote sensing. *GeoJournal* 89(1):1–28. <https://doi.org/10.1007/s10708-024-11005-w>
- Li MC, Feng SX (2020) Digital inclusive finance and urban-rural income gap: An analysis based on literature. *Contemp Econ Manag* 42(10):84–91. <https://doi.org/10.13253/j.cnki.ddjjgl.2020.10.011>
- Li P, Guo T (2024) Is digital finance a powerful means for Chinese cities to reduce environmental pollution in the fourth industrial revolution? *Technol Anal Strateg Manag* 36(10):2332–2351. <https://doi.org/10.1080/09537325.2022.2137398>
- Li Y, Yang X, Ran Q, Wu H, Irfan M, Ahmad M (2021) Energy structure, digital economy, and carbon emissions: evidence from China. *Environ Sci Pollut Res* 28(45):64606–64629. <https://doi.org/10.1007/s11356-021-15304-4>
- Li F, Wu Y, Liu J, Zhong S (2022a) Does digital inclusive finance promote industrial transformation? New evidence from 115 resource-based cities in China. *PLoS ONE* 17(8):e0273680. <https://doi.org/10.1371/journal.pone.0273680>
- Li Y, Wang M, Liao G, Wang J (2022b) Spatial spillover effect and threshold effect of digital financial inclusion on farmers' income growth—based on provincial data of China. *Sustainability* 14(3):1838. <https://doi.org/10.3390/su14031838>
- Lin B, Ma R (2022) How does digital finance influence green technology innovation in China? Evidence from the financing constraints perspective. *J Environ Manage* 320:115833. <https://doi.org/10.1016/j.jenvman.2022.115833>
- Liu Y, Chen L (2022) The impact of digital finance on green innovation: resource effect and information effect. *Environ Sci Pollut Res* 29(57):86771–86795. <https://doi.org/10.1007/s11356-022-21802-w>
- Liu R, Kang Y, Zhang J (2021a) Ideological taboos, entry barriers, and FDI attraction: evidence from China. *J Asian Econ* 76:101365. <https://doi.org/10.1016/j.asieco.2021.101365>
- Liu Y, Luan L, Wu W, Zhang Z, Hsu Y (2021b) Can digital financial inclusion promote China's economic growth? *Int Rev Financ Anal* 78:101889. <https://doi.org/10.1016/j.irfa.2021.101889>
- Liu D, Xie Y, Hafeez M, Usman A (2022) The trade-off between economic performance and environmental quality: does financial inclusion matter for emerging Asian economies? *Environ Sci Pollut Res* 29(20):29746–29755. <https://doi.org/10.1007/s11356-021-17755-1>
- Liu F, Yang S, Zhou Y, Huo W, Zhu L (2024) Does digital financial inclusion have an impact on high-quality development of trade? Evidence from China. *J Knowl Econ*. <https://doi.org/10.1007/s13132-024-02015-7>
- Lu Z, Wu J, Li H, Nguyen DK (2022) Local bank, digital financial inclusion and SME financing constraints: empirical evidence from China. *Emerg Mark Financ Trade* 58(6):1712–1725. <https://doi.org/10.1080/1540496X.2021.1923477>
- Lv M, Bai M (2021) Evaluation of China's carbon emission trading policy from corporate innovation. *Financ Res Lett* 39:101565. <https://doi.org/10.1016/j.frl.2020.101565>
- Lv C, Song J, Lee CC (2022) Can digital finance narrow the regional disparities in the quality of economic growth? Evidence from China. *Econ Anal Policy* 76:502–521. <https://doi.org/10.1016/j.eap.2022.08.022>
- Ma J, Li Z (2021) Does digital financial inclusion affect agricultural eco-efficiency? A case study on China. *Agronomy* 11(10):1949. <https://doi.org/10.3390/agronomy11101949>
- Ma XW, Ye Y, Shi XQ, Zou LL (2016) Decoupling economic growth from CO2 emissions: a decomposition analysis of China's household energy consumption. *Adv Clim Chang Res* 7(3):192–200. <https://doi.org/10.1016/j.jacce.2016.09.004>
- Mohieldin M, Hussein K, Rostom A (2019) On financial development and economic growth in Egypt. *J Hum Appl Social Sci* 1(2):70–86. <https://doi.org/10.1108/JHASS-08-2019-0027>
- Musah M, Kong Y, Vo XV (2021) Predictors of carbon emissions: an empirical evidence from NAFTA countries. *Environ Sci Pollut Res* 28(9):11205–11223. <https://doi.org/10.1007/s11356-020-11197-x>
- Niranjanamurthy M, Nithya BN, Jagannatha SJCC (2019) Analysis of Blockchain technology: pros, cons and SWOT. *Clust Comput* 22:14743–14757. <https://doi.org/10.1007/s10586-018-2387-5>
- Ogundari K, Awokuse T (2018) Human capital contribution to economic growth in Sub-Saharan Africa: does health status matter more than education? *Econ Anal Policy* 58:131–140. <https://doi.org/10.1016/j.eap.2018.02.001>
- Oketch MO (2006) Determinants of human capital formation and economic growth of African countries. *Econ Educ Rev* 25(5):554–564. <https://doi.org/10.1016/j.econedurev.2005.07.003>
- Onuoha FC, Dimnwobi SK, Okere KI, Ekesiobi C (2023) Funding the green transition: governance quality, public debt, and renewable energy consumption in Sub-Saharan Africa. *Utilities Policy* 82:101574. <https://doi.org/10.1016/j.jup.2023.101574>
- Ozili PK (2018) Impact of digital finance on financial inclusion and stability. *Borsa Istanbul Rev* 18(4):329–340. <https://doi.org/10.1016/j.bir.2017.12.003>
- Ozili PK, Lay SH, Syed AA (2023) Impact of financial inclusion on economic growth in secular and religious countries. *J Financ Regul Compliance*. <https://doi.org/10.2139/ssrn.4338410>
- Ozturk I, Ullah S (2022) Does digital financial inclusion matter for economic growth and environmental sustainability in OBRI economies? An empirical analysis. *Resour Conserv Recycl* 185:106489. <https://doi.org/10.1016/j.resconrec.2022.106489>
- Patel R, Sharma A (2020) Integration of the digital economy into India's data security Law. *J Econ Growth* 12(3):167–182
- Persyn D, Westerlund J (2008) Error-correction-based cointegration tests for panel data. *Stand Genomic Sci* 8(2):232–241. <https://doi.org/10.1177/1536867X0800800205>
- Pesaran MH (2004) General Diagnostic Tests for Cross Section Dependence in Panels, CESifo Working Paper Series No. 1229; IZA Discussion Paper No. 1240. Available at SSRN: <http://ssrn.com/abstract=572504>
- Pesaran MH (2007) A simple panel unit root test in the presence of cross-section dependence. *J Appl Econom* 22(2):265–312. <https://doi.org/10.1002/jae.951>
- Pesaran MH, Yamagata T (2008) Testing slope homogeneity in large panels. *J Econom* 142(1):50–93. <https://doi.org/10.1016/j.jeconom.2007.05.010>

- Rajan RG (2009) The credit crisis and cycle-proof regulation. *Fed Reserve Bank St Louis Rev* 91(5):397–402
- Rapih S, Wahyono B (2023) The relative impact of traditional and digital financial inclusion on economic growth: a threshold regression-based comparative analysis. *Int Rev Appl Econ* 37(6):781–803. <https://doi.org/10.1080/02692171.2023.2281460>
- Ren X, Zeng G, Gozgor G (2023a) How does digital finance affect industrial structure upgrading? Evidence from Chinese prefecture-level cities. *J Environ Manage* 330:117125. <https://doi.org/10.1016/j.jenvman.2022.117125>
- Ren X, Zeng G, Zhao Y (2023b) Digital finance and corporate ESG performance: empirical evidence from listed companies in China. *Pac Basin Financ J* 79:102019. <https://doi.org/10.1016/j.pacfin.2023.102019>
- Sadorsky P (2010) The impact of financial development on energy consumption in emerging economies. *Energy Policy* 38(5):2528–2535. <https://doi.org/10.1016/j.enpol.2009.12.048>
- Sarma M, Pais J (2011) Financial inclusion and development. *J Int Dev* 23(5):613–628. <https://doi.org/10.1002/jid.1698>
- Schumpeter JA (1912) *The theory of economic development*. Harvard University Press, Cambridge
- Sethi D, Acharya D (2018) Financial inclusion and economic growth linkage: some cross-country evidence. *J Financ Econ Policy* 10(3):369–385. <https://doi.org/10.1108/JFEP-11-2016-0073>
- Shabir M (2024) Does financial inclusion promote environmental sustainability: analyzing the role of technological innovation and economic globalization. *J Knowl Econ* 15(1):19–46. <https://doi.org/10.1007/s13132-022-01035-5>
- Shahbaz M, Hye QMA, Tiwari AK, Leita NC (2013a) Economic growth, energy consumption, financial development, international trade and CO2 emissions in Indonesia. *Renewable Sustain Energy Rev* 25:109–121. <https://doi.org/10.1016/j.rser.2013.04.009>
- Shahbaz M, Solarin SA, Mahmood H, Arouri M (2013b) Does financial development reduce CO2 emissions in Malaysian economy? A time series analysis. *Econ Model* 35:145–152. <https://doi.org/10.1016/j.econmod.2013.06.037>
- Shahbaz M, Li J, Dong X, Dong K (2022) How financial inclusion affects the collaborative reduction of pollutant and carbon emissions: the case of China. *Energy Econ* 107:105847. <https://doi.org/10.1016/j.eneco.2022.105847>
- Shao HH, Cheng JX, Wang YS, Li XM (2022) Can digital finance promote comprehensive carbon emission performance? Evidence from Chinese cities. *Int J Environ Res Public Health* 19:10255. <https://doi.org/10.3390/ijerph191610255>
- Shim Y, Shin DH (2016) Analyzing China's fintech industry from the perspective of actor-network theory. *Telecommun Policy* 40(2–3):168–181. <https://doi.org/10.1016/j.telpol.2015.11.005>
- Sun Y, Tang X (2022) The impact of digital inclusive finance on sustainable economic growth in China. *Financ Res Lett* 50:103234. <https://doi.org/10.1016/j.frl.2022.103234>
- Suri T, Bharadwaj P, Jack W (2021) Fintech and household resilience to shocks: evidence from digital loans in Kenya. *J Dev Econ* 153:102697. <https://doi.org/10.1016/j.jdevco.2021.102697>
- Swamy PA (1970) Efficient inference in a random coefficient regression model. *Econometrica* 38(2):311–323. <https://doi.org/10.2307/1913012>
- Tackie EA, Chen H, Ahakwa I, Atingabilli S, Ansah KA, Baku R (2020) Integration of economic, educational and socio-cultural capabilities for rural poverty alleviation in Northern Ghana. *Int J Trend Sci Res Dev* 5(1):528–535
- Tamazian A, Chousa JP, Vadlamannati KC (2009) Does higher economic and financial development lead to environmental degradation: evidence from BRIC countries. *Energy Policy* 37(1):246–253. <https://doi.org/10.1016/j.enpol.2008.08.025>
- Tang HL, Liu JM, Mao J, Wu JG (2020) The effects of emission trading system on corporate innovation and productivity—empirical evidence from China's SO2 emission trading system. *Environ Sci Pollut Res* 27(17):21604–21620. <https://doi.org/10.1007/s11356-020-08566-x>
- Thompson BS (2017) Can financial technology innovate benefit distribution in payments for ecosystem services and REDD+? *Ecol Econ* 139:150–157. <https://doi.org/10.1016/j.ecolecon.2017.04.008>
- Wan JY, Zhou Q, Xiao Y (2020) Digital finance, financing constraints and enterprise innovation. *Econ Rev* 1:71–83
- Wang X, Yang W, Ren X, Lu Z (2023a) Can financial inclusion affect energy poverty in China? Evidence from a spatial econometric analysis. *Int Rev Econ Financ* 85:255–269. <https://doi.org/10.1016/j.jref.2023.01.020>
- Wang X, Zhu Y, Ren X, Gozgor G (2023b) The impact of digital inclusive finance on the spatial convergence of the green total factor productivity in the Chinese cities. *Appl Econ* 55(42):4871–4889. <https://doi.org/10.1080/00036846.2022.2131721>
- Wang Z, Cao X, Ren X, Gozgor G (2024) Digital finance and the energy transition: evidence from Chinese prefecture-level cities. *Global Financ J* 61:100987. <https://doi.org/10.1016/j.gfj.2024.100987>
- Wen H, Yue J, Li J, Xiu X, Zhong S (2022) Can digital finance reduce industrial pollution? New evidence from 260 cities in China. *PLoS ONE* 17(4):e0266564. <https://doi.org/10.1371/journal.pone.0266564>
- White H (1980) A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*. <https://doi.org/10.2307/1912934>
- Wu S, Wang Z, Su C, Zhang W (2023) Digital media and green development path in asia: does digital financial inclusion matter? *Sustainability* 15(14):11359. <https://doi.org/10.3390/su151411359>
- Xi W, Wang Y (2023) Digital financial inclusion and quality of economic growth. *Heliyon* 9(9):e19731. <https://doi.org/10.1016/j.heliyon.2023.e19731>
- Xiao Y, Ma D, Zhang F, Zhao N, Wang L, Guo Z et al (2023) Spatiotemporal differentiation of carbon emission efficiency and influencing factors: from the perspective of 136 countries. *Sci Total Environ* 879:163032. <https://doi.org/10.1016/j.scitotenv.2023.163032>
- Xiong M, Li W, Teo BSX, Othman J (2022) Can China's digital inclusive finance alleviate rural poverty? An empirical analysis from the perspective of regional economic development and an income gap. *Sustainability* 14(24):16984. <https://doi.org/10.3390/su142416984>
- Xu S, Wang J (2023) The impact of digital financial inclusion on the level of agricultural output. *Sustainability* 15(5):4138. <https://doi.org/10.3390/su15054138>
- Xue L, Zhang X (2022) Can digital financial inclusion promote green innovation in heavily polluting companies? *Int J Environ Res Public Health* 19(12):7323. <https://doi.org/10.3390/ijerph19127323>
- Yang C, Masron TA (2022) Impact of digital finance on energy efficiency in the context of green sustainable development. *Sustainability* 14(18):11250. <https://doi.org/10.3390/su141811250>

- Yang L, Zhang Y (2020) Digital financial inclusion and sustainable growth of small and micro enterprises—evidence based on China's new third board market listed companies. *Sustainability* 12(9):3733. <https://doi.org/10.3390/su12093733>
- Yang B, Ma F, Deng W, Pi Y (2022) Digital inclusive finance and rural household subsistence consumption in China. *Economic Anal Policy* 76:627–642. <https://doi.org/10.1016/j.eap.2022.09.007>
- Yang A, Huan X, Teo BSX, Li W (2023a) Has green finance improved China's ecological and livable environment? *Environ Sci Pollut Res* 30(16):45951–45965. <https://doi.org/10.1007/s11356-023-25484-w>
- Yang M, Chen H, Long R, Yang J (2023b) How does government regulation shape residents' green consumption behavior? A multi-agent simulation considering environmental values and social interaction. *J Environ Manage* 331:117231
- Yin Z, Gong X, Guo P, Wu T (2019) What drives entrepreneurship in digital economy? Evidence from China. *Econ Model* 82:66–73. <https://doi.org/10.1016/j.econmod.2019.09.026>
- Yu N, Wang Y (2021) Can digital inclusive finance narrow the Chinese urban–rural income gap? The perspective of the regional urban–rural income structure. *Sustainability* 13(11):6427. <https://doi.org/10.3390/su13116427>
- Yu L, Zhao D, Xue Z, Gao Y (2020) Research on the use of digital finance and the adoption of green control techniques by family farms in China. *Technol Soc* 62:101323. <https://doi.org/10.1016/j.techsoc.2020.101323>
- Zavolokina L, Dolata M, Schwabe G (2016) The FinTech phenomenon: antecedents of financial innovation perceived by the popular press. *Financ Innov* 2(1):1–16. <https://doi.org/10.1186/s40854-016-0036-7>
- Zhang YJ (2011) The impact of financial development on carbon emissions: an empirical analysis in China. *Energy Policy* 39(4):2197–2203. <https://doi.org/10.1016/j.enpol.2011.02.026>
- Zhang W, Xu H (2017) Exploring the causal relationship between carbon emissions and land urbanization quality in China using a panel data analysis. *Environ Dev Sustain* 19(4):1445–1462. <https://doi.org/10.1007/s10668-016-9813-8>
- Zhang Q, Li J, Li Y, Huang H (2023a) Coupling analysis and driving factors between carbon emission intensity and high-quality economic development: evidence from the Yellow River Basin China. *J Clean Prod* 423:138831. <https://doi.org/10.1016/j.jclepro.2023.138831>
- Zhang R, Wu K, Cao Y, Sun H (2023b) Digital inclusive finance and consumption-based embodied carbon emissions: a dual perspective of consumption and industry upgrading. *J Environ Manage* 325:116632. <https://doi.org/10.1016/j.jenvman.2022.116632>
- Zhang S, Han Z, Guo M (2024) FDI, new development philosophy and China's high-quality economic development. *Environ Dev Sustain* 26(10):25227–25255. <https://doi.org/10.1007/s10668-023-03677-0>
- Zheng H, Li X (2022) The impact of digital financial inclusion on carbon dioxide emissions: empirical evidence from Chinese provinces data. *Energy Rep* 8:9431–9440. <https://doi.org/10.1016/j.egy.2022.07.050>
- Zheng S, Sheng B, Ghafoor A, Ashraf AA, Qamri GM (2023) Investigating the environmental externalities of digital financial inclusion and the COVID-19 pandemic: an environmental sustainability perspective. *Environ Sci Pollut Res* 30(33):80758–80767. <https://doi.org/10.1007/s11356-023-27433-z>
- Zhou Z, Zhang Y, Yan Z (2022) Will digital financial inclusion increase Chinese farmers' willingness to adopt agricultural technology? *Agriculture* 12(10):1514. <https://doi.org/10.3390/agriculture12101514>
- Zhu Y, Lin B (2004) Sustainable housing and urban construction in China. *Energy Build* 36(12):1287–1297. <https://doi.org/10.1016/j.enbuild.2003.11.007>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.